

Shock Propagation through Migrant Networks: Evidence from Subnational Migration in the Ecuador-Spain Migration Corridor

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Abstract

International migration entails human and financial flows that can expose low-income communities to foreign economic shocks. This paper studies the Ecuador-Spain corridor during the Global Financial Crisis to investigate the impact of foreign labor market shocks on migrant origin communities. I leverage international survey and census data to craft an emigrant shift-share instrument estimating subnational Ecuadorian exposure to Spanish regional labor market downturns. Spanish shocks propagate to Ecuador, with more exposed Ecuadorian regions experiencing significant increases in return migration and reductions in remittance inflows. Spanish labor demand shocks impact Ecuadorian labor markets, as employment rates increase for women and older segments of the population, wages fall for less educated workers, self-employment increases and male construction employment declines.

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I accept full responsibility for the accuracy or reliability of my calculations based on Spanish data from Spain's Instituto Nacional de Estadística (INE).

1 Introduction

The study of international migration increasingly addresses economic implications for migrant sending regions. Despite the distances traveled, migrants often maintain connections with those left behind, and many who set out intend to return in the future (Dustmann and Mestres 2010; Dustmann and Goerlach 2016). Migration thus creates connections between disparate countries, leaving imprints on both. An extensive literature investigates the impact of immigration on destination countries, particularly labor markets in the developed world (Card 1990; Ottaviano and Peri 2012; Dustmann, Schönberg, and Stuhler 2016; Edo 2020). This paper contributes to a growing body of work (Mishra 2007; Dustmann, Frattini, and Rosso 2015; Salem and Seck 2025) investigating how emigration affects the places emigrants leave.

Emigration has various impacts on migrant origin regions, which the literature has begun to explore. Emigration may prompt further migration (Spitzer and Zimran 2023), increase wages (Mishra 2007; Dustmann, Frattini, and Rosso 2015) and economic development (Salem and Seck 2025), and spur large financial flows in the form of migrant remittances (Yang 2011; Joseph, Nyarko, and Wang 2018; Clemens and McKenzie 2018). Migrant networks may also expose low-income communities to foreign labor markets and business cycles in emigrant destinations. Shocks could propagate through remittances and return migration, with theoretically ambiguous net effects. While one would expect a remittance shock to reduce incomes (Khanna et al. 2022), return migration could either amplify this shock by putting downward pressure on wages (Caballero, Cadena, and Kovak 2023), or counteract it if returnees stimulate local labor demand, for example through entrepreneurship and the adoption of technologies learned in the destination (Hausmann and Nedelkoska 2018).

I study the Ecuador-Spain migration corridor to investigate how foreign labor market shocks can propagate through migrant networks and their ultimate impact on migrant sending regions. Nearly half a million Ecuadorians had migrated to Spain by 2005, equivalent to 3.3% of Ecuador’s population and 1.1% of Spain’s population (UN 2020; Carioli et al. 2024; INE 2023), and remittances from Spain to Ecuador totaled 2.7% of Ecuadorian GDP in 2007 (Banco Central del Ecuador 2025; WBG 2025a). In the aftermath of the Global Financial Crisis, Spain experienced a recession as its construction industry imploded (Pardo 2011; Roig and Recaño Valverde 2012; Bentolila et al. 2012). I find that shockwaves from this European recession transmitted to South America through migrants. Spanish employment shocks reduced immigrant employment and spurred a decline in remittances and an increase in return migration to Ecuador, with implications for Ecuadorian local labor markets.

An advantage of the Ecuador-Spain context is that shock propagation would primarily occur through migrant networks rather than trade. Gravity models hold that physical distance is an important determinant of flows of labor and goods (Anderson 2011; Beine, Bertoli, and Fernández-Huertas Moraga 2016). True to this prediction, the highly studied Mexico-US migration corridor observed large flows of goods as well as labor: the United States was the largest export market for Mexico in 2007, receiving 82% of Mexican exports (UN 2026). However, Ecuador and Spain are quite distant, sharing neither borders, a continent, nor an ocean, so a standard gravity model would predict minimal trade or migration between these countries. While Spanish migration policy and a combination of strong push

and pull factors provoked mass Ecuadorian migration to Spain in the early 21st century (see Section 3.2), high levels of migration did not beget equally high levels of trade. Spain hosted 46% of Ecuadorian emigrants in 2005 (UN 2020), but only imported 2.4% of Ecuadorian exports in 2007 (UN 2026). This imbalance meant the Spanish recession would primarily impact Ecuador through migrant job loss, rather than a demand shock for traded goods. The Ecuador-Spain context permits me to analyze international shock propagation through migrant networks without the need to account for simultaneous propagation through trade.

For my empirical methodology I design an emigrant employment shift-share instrument to estimate how exposure to Spanish labor market downturns impacted development in Ecuadorian regions. I estimate subnational Ecuadorian exposure to regional Spanish labor market downturns, using a Spanish shift-share instrument based on employment in migrant industries to predict migrant labor shocks. These predictions form the shifts of the emigrant shift-share instrument, while the subnational migration network provides the shares. I estimate a subnational migrant matrix for Ecuadorians in Spain from a pre-crisis nationally representative Spanish immigrant survey and Ecuadorian census emigration microdata (INE 2008b; INEC 2026a; INEC 2026b). I instrument for employment changes in Spain with Spanish labor force survey data, and measure outcomes in Ecuador using Ecuadorian labor force survey and Ecuadorian Central Bank data (INE 2024; INEC; 1990–2019; Banco Central del Ecuador 2025). My emigration employment shift-share instrument resembles that of Caballero, Cadena, and Kovak (2023); I innovate by instrumenting for migrant employment shifts.

Regression results indicate shock propagation through both remittance and return migration channels. Results suggest impacted households experienced financial strain, as full-time employment rates rose for older female cohorts, and wages for less educated men and women fell for cohorts in Ecuadorian cantons most exposed to Spanish labor market downturns, relative to less exposed cantons. In addition to increasing their supply of labor, households may have reduced housing investments, as male construction employment rates in affected Ecuadorian regions fell significantly in the short-run. It appears that when emigrants in Spain lost their construction jobs, so did their peers back home.

I contribute to economics literature on migration, development, and labor, particularly development impacts of shocks to emigrants. I also contribute to literature on Ecuador and migrants in the Spanish labor market.

This paper addresses economics literature on migration (Borjas 1987; Chiquiar and Hanson 2005), migrant networks (Munshi 2003; Spitzer and Zimran 2023), remittances (Osili 2004; Yang 2011; Clemens and McKenzie 2018; Joseph, Nyarko, and Wang 2018; Khanna et al. 2022), and temporary and return migration (Dustmann and Goerlach 2016; Dustmann and Mestres 2010; Adda, Dustmann, and Görlach 2022). It relates to return migration impacts on development (Bahar et al. 2024; Bucheli and Fontenla 2025; Wahba 2026), particularly when stimulated by shocks in the migrant-destination (Hausmann and Nedelkoska 2018; Thomas Pearson 2025; Caballero, Cadena, and Kovak 2023). The paper builds on a literature investigating emigration impacts on economic development and labor markets (Gibson, McKenzie, and Stillman 2011; Mishra 2007; Borjas 2008; Aydemir and Borjas 2007; Akram, Chowdhury, and Mobarak 2017; Kosack 2021; Salem and Seck 2025), in particular the impact of labor market shocks faced by emigrants (Caballero, Cadena, and Kovak 2023; Gröger 2021).

The paper relates to literature about Ecuador, Spain, and the global financial crisis focused on labor and migration. It builds on social science literature on Ecuadorian labor markets (Gallardo and Ñopo 2009; Ontaneda Jiménez et al. 2022; Ontaneda Jiménez et al. 2024; Meneses et al. 2026), Ecuadorian emigration (Jokisch and Kyle 2008; Bertoli, Fernández-Huertas Moraga, and F. Ortega 2011; Soruco, Piani, and Rossi 2008), and Ecuadorian return migration (Perraudin 2017; Mercier et al. 2016). The paper contributes to labor economics research on Spanish and European labor markets, including migrant labor in Spain (Fernández and C. Ortega 2008; Bentolila, Dolado, and Jimeno 2008; González and F. Ortega 2011; Roig and Recaño Valverde 2012; Özgüzel 2021), Spain’s dual labor market (Dolado, García-Serrano, and Jimeno 2002; Bentolila et al. 2012), European labor market responses to economic shocks (Hennig 2023; Lechler 2019), and European migrant responses to shocks (Mitze 2019; Basso, D’Amuri, and Peri 2019).

Methodologically, I build on the economic literature using shift-share instruments to study labor markets and migration. It builds on literature using industry employment shift-share instruments (Bartik 1991; Charles, Hurst, and Schwartz 2018; Goldsmith-Pinkham, Sorkin, and Swift 2020), particularly in the European context (Lechler 2019; Mitze 2019; Basso, D’Amuri, and Peri 2019). The paper’s empirical strategy builds on a growing literature crafting emigrant shift-share instruments for causal inference (Caballero, Cadena, and Kovak 2023; Gröger 2021; Godlonton and Theoharides 2025).

The paper is organized as follows: after describing data in Section 2 and the Ecuadorian-Spanish migration context in Section 3, I outline the empirical strategy in Section 4. Section 5 describes results, and Section 6 concludes.

2 Data

2.1 Data Sources

My empirical strategy relies on an emigrant employment shift-share instrument, with emigration rates constituting the shares and instrumented Spanish employment changes representing the shifts. Ecuadorian census and Spanish survey microdata allow me to estimate a subnational international migrant network for the Ecuador-Spain migration corridor in 2007, from which I derive subnational emigration rates. Spanish labor force survey microdata permits the estimation and instrumentation of Spanish regional employment shifts. For intermediate outcomes, I consult Ecuadorian Central Bank remittance flow data and Ecuadorian labor force survey microdata to track changes in remittances and return migration rates at the Ecuadorian province and canton levels. I standardize Ecuadorian geographies to historic province and canton boundaries (see Section A for details). Ecuadorian labor force survey microdata enables me to compute labor market outcomes in Ecuador, including local labor market employment rates and wages by age, gender, and education cohort. Sources of data include:

Encuesta de Población Activa (EPA), Spain: I use Spanish quarterly labor force survey (Encuesta de Población Activa, henceforth EPA) public use microdata files from 2005-2019 (INE 2024), produced by Spain’s Instituto Nacional de Estadística (INE). I calculate annual employment rates by restricting to second quarter labor force survey microdata, similar to Fernández and C. Ortega (2008) and Dolado, García-Serrano, and Jimeno (2002).¹ Public use EPA data records employment, 1-digit economic activity and occupation, contract type, nativity, geographic area (e.g. continent) of birth for the foreign born, Spanish province of residence, and demographic characteristics.

Encuesta Nacional de Empleo, Desempleo y Subempleo (ENEMDU), Ecuador: I standardize and analyze labor force survey microdata from Ecuador’s Encuesta Nacional de Empleo, Desempleo y Subempleo (ENEMDU) from 1990-2019 (INEC; 1990–2019), produced by Ecuador’s Instituto Nacional de Estadística y Censos (INEC). ENEMDU records detailed geographic, demographic, labor market, and income data, as well as housing and asset data starting in 2007. My main analysis uses nationally representative annual or December surveys from 2000-2019, excluding Galapagos, and dropping 2002 data, similar to Ontaneda Jiménez et al. (2022) and Ontaneda Jiménez et al. (2024).² I compute real monetary values using US CPI, deflating to real 2015 USD, and convert currencies to USD using World Bank exchange rates (OECD 2025; WBG 2025b).

Spanish Immigrants: A nationally representative Spanish immigrant survey in 2007 records detailed demographic, remittance, labor market, and geographic information (INE 2008b). Nativity is recorded in Spanish censuses, including 5% sample microdata for 1981,

¹Spanish labor force survey respondents are sampled for six consecutive quarters, with a sixth of the sample replaced each quarter (Fernández and C. Ortega 2008).

²Ontaneda Jiménez et al. (2022) and Ontaneda Jiménez et al. (2024) drop the 2002 ENEMDU since that year’s survey only had an urban sample.

1991 and 2001, and 10% sample microdata for the 2011 census from iPUMS International (Ruggles et al. 2025a; INE n.d.[b]; INE n.d.[c]; INE n.d.[d]; INE n.d.[a]), and full-census tabulations from the 2001 census note birth country and residence in Spain (INE 2004). Spanish population registry (*Estadística Padron Continuo*) full-count microdata and tabulations from 2003-2022 record simple demographic information, including birth-country and residence, for Spanish residents at an annual frequency (INE 2022b; INE 2023).

Ecuadorian Census: Public-access, geographically-standardized full-count Ecuadorian census microdata for 2001, 2011, and 2022 includes data on emigrants as well as demographic, labor, and household data for Ecuadorian residents (INEC 2026a; INEC 2026b; INEC 2026c). I also consult 10% sample Ecuadorian census microdata for 2010 from iPUMS international (Ruggles et al. 2025a; INEC n.d.), and INEC census tabulations for the 2001, 2011, and 2022 censuses (INEC; 2003–2011; INEC 2025a; INEC 2025b).

Migration Flows: INEC 2017 Anuario of International transit statistics record total annual departures and arrivals between Ecuador and Spain from 1997-2017 (INEC 2018). The registry of Spaniards Resident Abroad counts Spanish citizens living in Ecuador at a yearly basis starting in 2009, recording the number born in Ecuador (INE 2026a). ENEMDU surveys often record prior location of residence, including foreign countries, by which I proxy return migration (INEC; 1990–2019).

Remittance Flows: The Ecuadorian Central Bank publishes detailed remittance inflow and outflow tables for total subnational flows and internationally-directional national flows at quarterly and annual frequencies from 2007 onwards (Banco Central del Ecuador 2025), which I convert to real values using Ecuadorian Central Bank price index data (INEC 2025c). ENEMDU surveys generally record prior month remittance receipt and income (INEC; 1990–2019).

International Migration: The United Nations estimates a global migrant matrix with 5 year estimates of binational migrant stocks (UN 2020), and I consult ACS microdata from iPUMS USA to investigate Ecuadorian immigrant outcomes in the United States (U.S. Census Bureau 2007–2011; Ruggles et al. 2025b).

Ecuadorian Regional Economic Data: The Ecuadorian Central Bank provides canton-level sectoral annual production figures from 2007-2019 (ECB 2026), which I incorporate in robustness checks after converting to real 2014 USD using Ecuadorian Central Bank price index data (INEC 2025c).

2.2 Subnational Migration Matrix

I estimate subnational migration networks from Ecuador to Spain using the 2007 Spanish migrant survey and Ecuadorian census microdata from 2001 and 2010 (INE 2008b; INEC 2026a; INEC 2026b). The 2007 Spanish migrant survey records both the Spanish province of residence and Ecuadorian province of birth for Ecuadorian respondents, and I use these variables to estimate a pre-crisis province-to-province migration matrix for migration from Ecuador to Spain (INE 2008b). Using survey responses and weights, I estimate $ImmigrSpain_{p,j,2007}$, the number of migrants from Ecuadorian province p living in Spanish province j in 2007. I combine this with Ecuadorian census emigration microdata to estimate a geographically finer canton-to-province migration matrix, from which I derive $\widehat{Emigr}_{c,p,j,2007}$, the number of emigrants from Ecuadorian canton c in Ecuadorian province p living in Spanish province j . I assign province-to-province immigrant stocks $ImmigrSpain_{p,j,2007}$ to origin-cantons c based on $\frac{EmigrCensus_{c,p,1995-2007}}{EmigrCensus_{p,1995-2007}}$, canton c 's share of total province p emigration to Spain between 1995-2007 as recorded in the Ecuadorian census (INEC 2026a; INEC 2026b). I choose to estimate migration stocks using the Spanish immigrant survey instead of the Ecuadorian census because recency bias and household migration may lead the census to undercount migration.³ This method assumes that migrants from cantons in the same Ecuadorian province have identical subnational distributions within Spain, as so:

$$\widehat{Emigr}_{c,p,j,2007} = ImmigrSpain_{p,j,2007} \cdot \frac{EmigrCensus_{c,p,1995-2007}}{EmigrCensus_{p,1995-2007}} \quad (1)$$

As my Ecuadorian labor outcomes are restricted to mainland regions, I focus on historic mainland provinces in Ecuador. The matrix covers 20 historic mainland Ecuadorian provinces and 45 out of 52 Spanish provinces. The destination provinces account for 99.5% of the Ecuadorian immigrant stock in Spain in 2007 according to the Spanish population registry (INE 2023). Figure 1 compares Ecuadorian province-level log emigration to Spain according to Ecuadorian censuses and the Spanish migrant survey (Figure 2 does the same for immigration to Spanish provinces). Reassuringly, although Spanish immigrant survey estimates are slightly greater than the Ecuadorian census on average, I find high correlation between both data sources, suggesting accurate responses to the Spanish migrant survey.

³Households may forget to mention household emigrants who migrated many years prior, leading to an undercount of emigrants. Entire household migration would not be recorded in the census as no one would remain in Ecuador to respond to a census enumerator; the census recorded emigration by asking households whether any household members had emigrated over a given period of time.

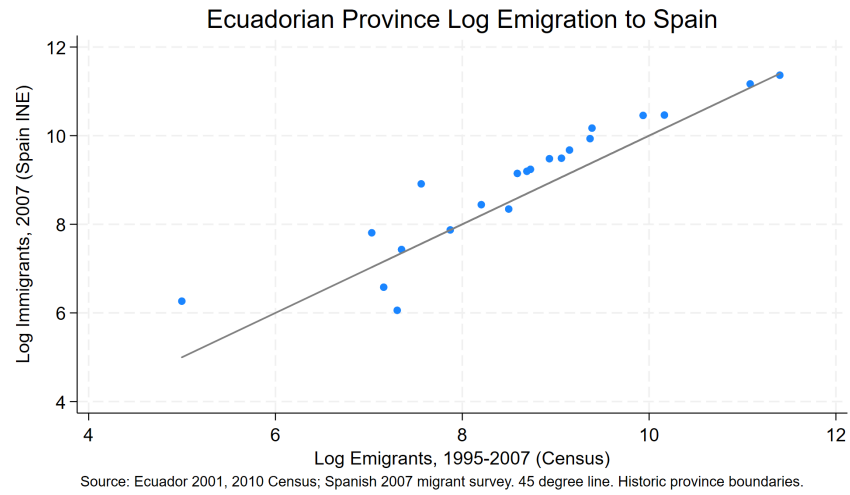


Figure 1: Subnational Emigration

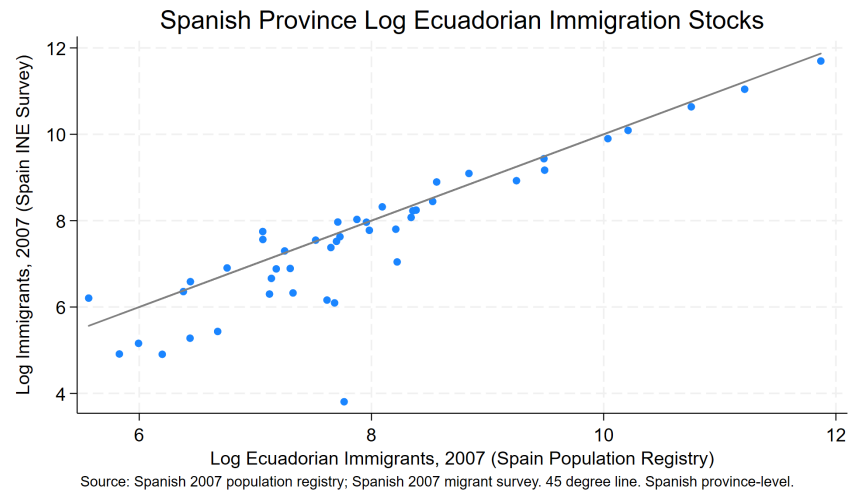


Figure 2: Subnational Immigration

3 Context

3.1 Country Settings

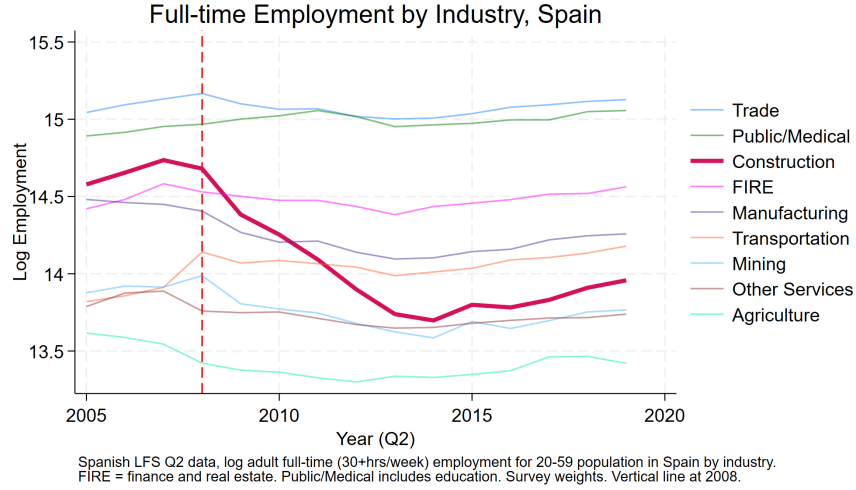
3.1.1 Spanish Labor Markets and Recession

Spanish labor market features turned Spain’s recession into a large negative employment shock with major implications for Spanish immigrants, presenting a clear example of how migration can expose developing communities to foreign risks. Spain experienced a nearly fifteen year expansion from 1993 to 2008, but economic growth gave way to two recessions following the Global Financial Crisis (Spanish Business Cycle Dating Committee 2026). These double recessions between 2008 and 2013 entailed a continuous decline in employment exacerbated by structural weaknesses in the Spanish labor market (Spanish Business Cycle Dating Committee 2026; INE 2024). Spain had experienced impressive growth while becoming a high-income country and Eurozone member (Calvo-Gonzalez 2021; Dolado 2017), but its economy was concentrated in a few vulnerable sectors (Pardo 2011) and built upon labor market institutions that created high employment volatility (Bentolila and Jimeno 2006; Bentolila et al. 2012). Spain had a dual labor market, with strong job protections for those working under indefinite contracts and weak protections for those under temporary contracts (Dolado, García-Serrano, and Jimeno 2002; Dolado 2017; Bentolila, Dolado, and Jimeno 2020), and collective bargaining that set wages and hours (Bentolila and Jimeno 2006), meaning employers would adjust labor costs at the extensive margin by laying off temporary contract workers instead of making adjustments at the intensive margin. Temporary contracts had covered a third of the labor force, and a majority of migrant workers (Bentolila et al. 2012; Fernández and C. Ortega 2008), yet full-time (30hrs/week+) temporary contract employment fell by 48% from 2007Q2-2013Q2 for prime age (20-59) adults (see Figure C.2) (INE 2024). Spanish employment was concentrated in construction and tourism (Pardo 2011), which accounted for most job loss from 2008-10; full-time prime age adult employment in construction fell by 63% between 2007Q2 to 2013Q2 (see Figure 3a) (INE 2024). Full-time employment loss over this period amounted to 3.31 million, 57.5% accounted for by temporary contract employment and 47.8% by construction employment.

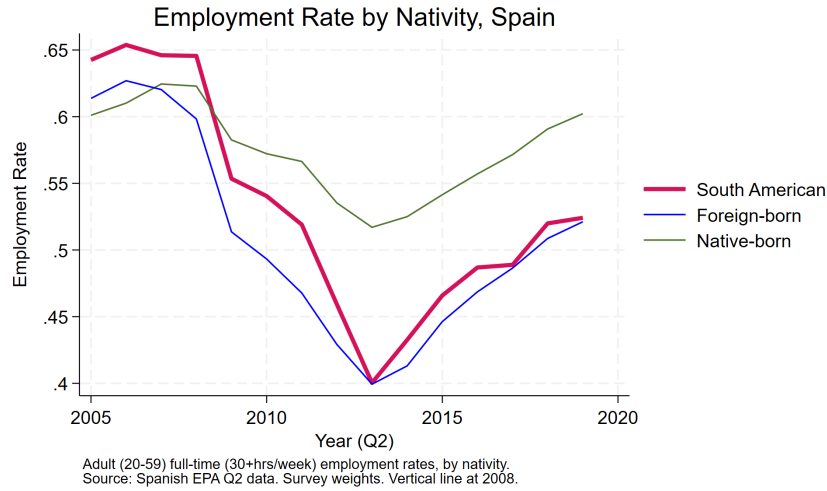
Migrant workers experienced an extreme version of these labor market fluctuations due to their position in the labor market. Migrants were more likely to work in the informal sector (Roig and Recaño Valverde 2012), under temporary contracts (Fernández and C. Ortega 2008), and in non-tradeable industries including construction (González and F. Ortega 2011), which was a strong contributor to migrant job loss (Roig and Recaño Valverde 2012). Although South American employment rates had exceeded the native-born pre-crisis, they dropped below natives in 2009 and did not catch up in the following decade (see Figure 3b) (INE 2024). Ecuadorian-born adult employment rates fell by nearly 20 percentage points for women and 30 percentage points for men to rates of about 50% (INE 2008b; INE n.d.[a]).

3.1.2 Ecuadorian Development

Ecuador experienced a financial crisis at the turn-of-the century, followed by economic growth in the early 2010s. Hyperinflation and a banking crisis in the late 1990s led to dollarization



(a) Spanish Employment by Industry



(b) Employment Rates by Nativity

Figure 3: Spanish Labor Market

in 2000 and created an impetus for emigration (Jokisch and Kyle 2008). Median urban household incomes fell sharply, but later rebounded (see Figure C.5a), and national median household incomes rose during the early 2010s (see Figure C.5b) as poverty rates fell. Oil prices and government expenditures likely contributed to the economic upswing. Oil was Ecuador's largest export (World Bank 2026), and Rafael Correa, elected president of Ecuador in 2007, leveraged favorable prices to increase state spending on infrastructure and social programs (De La Torre 2018). However, Ontaneda Jiménez et al. (2024) relate that Ecuador's economic expansion ended with contractions in 2015 and 2016, followed by lower growth from 2017-18.

The Ecuadorian labor market suffered during the national crisis in the 1990s but recovered by the mid-2000s (see Figures C.6a and C.6b). By 2007, 84% of prime-age men and 44% of prime-age women were full-time employed (INEC; 1990–2019). While the population

was largely urban, many adults had no more than a primary education. In 2007, 39.7% of prime-age (25-54) adults had ended in primary education, 34.4% in secondary education, and 21.5% in superior education, and 31% of full-time prime-age jobs were categorized as informal in 2013 (INEC; 1990–2019). Men and women mostly worked in market services. The Ecuadorian labor market thus had only moderate levels of human capital and a high degree of informality when its emigrants experienced Spanish employment shocks tied to the Global Financial Crisis. Nevertheless, improving domestic economic conditions simultaneous to the Spanish recession would have incentivized return migration.

3.2 There and Back Again: Ecuador-Spain Migration

3.2.1 Migration Policy

Ecuadorian migration to Spain occurred within a framework of *de facto* migrant-friendly mobility and integration policies. A visa-waiver program from 1963 to August 2003 allowed Ecuadorians to enter Spain as tourists;⁴ they could overstay to work and benefit from a series of amnesty programs to obtain *ex-post* regularization⁵ (Roig and Recaño Valverde 2012; Bertoli, Fernández-Huertas Moraga, and F. Ortega 2011; Jokisch and Kyle 2008). Latin America immigrants could naturalize following a two-year residency requirement, eight years less than other immigrants (Enríquez 2014).⁶ These Spanish policies facilitated migration and integration for nearly half a million Ecuadorians, as I describe in Section 3.2.2.

In addition to facilitating Ecuadorian migration to Spain, Spanish policy played a role in prompting some return migration. In the midst of its economic crisis, the Spanish state sought to incentivize out-migration through voluntary and involuntary means (Perraudin 2017). Based on official statistics, nearly seven thousand Ecuadorians participated in one of three voluntary return migration programs offered by the Spanish government – an unemployment benefits program (APRE), assistance for vulnerable migrants, and entrepreneurial return migration (see Figure C.3b) (ISSM 2026a; ISSM 2026b; ISSM 2026c). The APRE program, which allowed unemployed return migrants to cash out unemployment contributions, had the most uptake (ISSM 2026a), as did the vulnerable reintegration assistance scheme, which provided a one-way ticket home and can be used by undocumented migrants (Perez 2026; ISSM 2026b). Spain increased Ecuadorian deportations during the crisis to a similar order of magnitude as its voluntary return migration programs, as nearly five thousand return orders for Ecuadorians were fulfilled between 2008 and 2014 (Eurostat 2026b).

Ecuadorian policy also encouraged return migration. Perraudin (2017) notes that Ecuador allowed dual citizenship beginning in 1998, and that, in the wake of the crisis, Ecuadorians with formal Spanish paperwork were more likely to move back to Ecuador since the option to return to Spain remained open for them. Mercier et al. (2016) describes the creation of a secretariat of migration in 2007, which claimed to have aided over forty thousand Ecuadorians

⁴Ecuadorian flows fell sharply after Spain introduced a visa requirement (Bertoli, Fernández-Huertas Moraga, and F. Ortega 2011).

⁵Beginning in 1991, Spain offered multiple amnesty waves to regularize undocumented migrants; it codified a permanent regularization process into law in 2006 (Roig and Recaño Valverde 2012).

⁶About 62,000 Ecuadorians became naturalized Spanish citizens through residency from 1995–2007, and nearly 254,000 from 1995–2013 (see Figure C.3a) (OPI 1997; OPI 1999; OPI 2001; OPI and SEM 2026).

living abroad return migrate between 2008 and 2013.

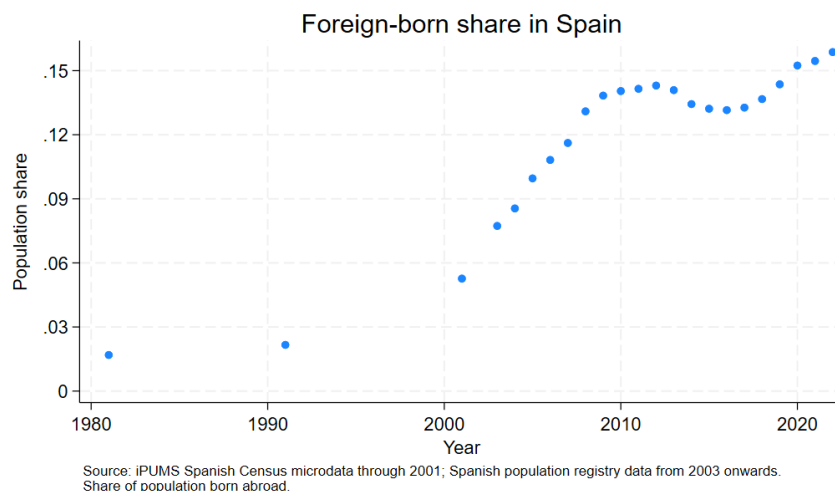


Figure 4: Spain foreign-born share

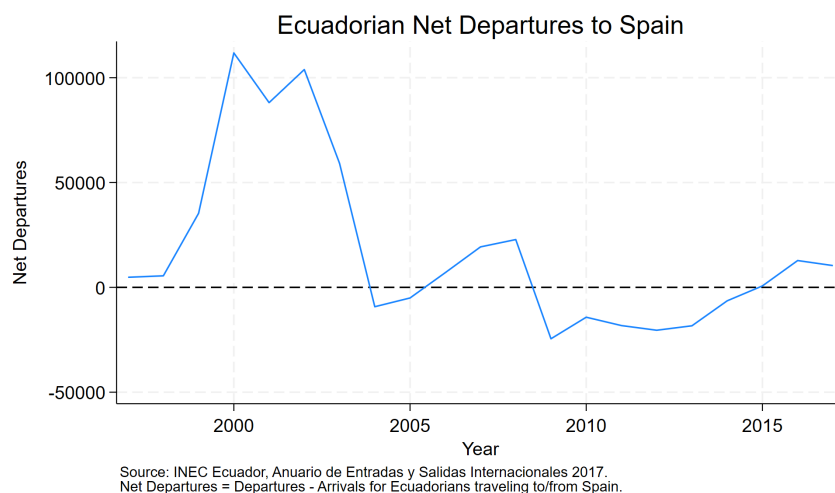


Figure 5: Ecuadorian Net Departures to Spain (INEC)

3.2.2 Migrant Flows

Spanish economic growth, migrant-friendly policies, and strong push factors led Ecuadorian migrants to become a major tributary of a growing river of migration to Spain. Spain's foreign-born population share rose fivefold over two decades, from 2.2% to 12.1% between 1991 and 2011 (see Figure 4) (INE n.d.[c]; INE n.d.[a]; Ruggles et al. 2025a).⁷ Economic push factors in Ecuador, including hyperinflation and a banking crisis (Jokisch and Kyle 2008),

⁷By 2005, Morocco, Ecuador, Romania, Colombia and Argentina were the five countries of origin with the largest immigrant stocks in Spain (UN 2020) (see Table D.1).

stimulated emigration, while *de facto* pro-migration policies facilitated Spanish migration and integration. Net departures of Ecuadorians to Spain from Ecuadorian transit data reveal a sharp rise in emigration starting in 1999, climbing to an average of nearly 100,000 net departures per year from 2000 to 2002 as seen in Figure 5 (INEC 2018). Nearly a half million Ecuadorians lived in Spain by 2005, becoming the second largest Spanish immigrant group (UN 2020). After Spain entered a recession, migrant employment dropped, and a subset of Ecuadorians opted to return.

Following the Spanish recession, migration flows slowed and reversed; while most migrants remained in Spain, some Ecuadorians returned to their home country. Spanish population registry counts suggest net declines in the Ecuadorian-born immigrant population of over 50,000 from 2010 to 2017, although the stock remained above 400,000 (INE 2022b). Ecuadorian transit data record persistently negative net flows of Ecuadorians to Spain from 2009-2014, indicative of a countercyclical return migration flow (INEC 2018). Return migration exceeds that directly sponsored by Spanish policy. Return migration for Ecuadorians with Spanish citizenship appears over twice as large as Spanish state-sponsored return migration. The number of Ecuadorian-born Spanish nationals residing in Ecuador, presumably return migrants ineligible to participate in Spanish state-sponsored return programs, increased by nearly twenty-five thousand between 2009 and 2015 (INE 2026a), more than double the Spanish state-sponsored return migration of about twelve thousand (59% voluntary) over the same period (ISSM 2026c; ISSM 2026b; ISSM 2026a; Eurostat 2026a). I investigate the attributes of Ecuadorian return migrants and how they differed from those who remained in Spain and those they rejoined in Ecuador, as this could shape their potential impacts on Ecuadorian labor markets.

3.3 Migrant Selection

An extensive literature documents and investigates migrant selection, differences in attributes between those who migrate and those who do not. Most work points to positive migrant selection, that is, to migrants possessing higher levels of human capital than non-migrants. Other work documents positive selection on health and culture. Differences between Ecuadorian emigrants and non-migrants have implications for this setting as they return migration

The labor market impact of the reversal in migration flows would depend on returnees' characteristics, which relate to the attributes of emigrants and any changes that occurred during migration episodes, such as the accumulation of human or financial capital. An extensive literature investigates migrant selection, aggregate differences between emigrants and non-migrants; most work points to positive selection for human capital (Borjas 1987; Chiquiar and Hanson 2005). Ecuadorian emigrants to Spain were moderately positively selected on education and negatively selected on age. I analyze migrant selection through descriptive regressions of microdata on adults aged 25-54 from national surveys conducted in Spain, the United States, and Ecuador in 2007. I pool Ecuadorian-born respondents from the 2007 Spanish Immigrant Survey and the 2007 American Community Survey with respondents in the 2007Q4 Ecuadorian labor force survey (ENEMDU) (U.S. Census Bureau 2007-2011; INE 2008a; INEC; 1990-2019).⁸ I test for selection along gender, age, education, household

⁸For the ACS, I exclude those born in Ecuador as US citizens. The 2007Q4 ENEMDU does not record

characteristics, and employment. Accounting for differences in age, migrants to both Spain and the United States are positively selected on having at most some secondary education; this share is more than double for Spanish migrants (31.1 percentage points higher). However, selection on college education differs by destination, with Spanish migrants being negatively selected on college education. Ecuadorian migrants to Spain are more educated than non-migrants, but this comes exclusively from having a greater share in secondary education.

Prior to the crisis, international migrants experience higher employment rates and earnings than non-migrants. Ecuadorians in Spain and the US are more likely to be employed than non-migrants (see Table D.3), and both are more likely to work in construction, reflecting employment in destination housing booms. Labor market earnings are an order of magnitude higher in both destination countries in comparison to Ecuador (see Table D.4). Average monthly wages are over three times higher in Spain than Ecuador, and nearly seven times higher in the US (Spanish returns even rise after accounting for education). Migrants experienced higher employment rates and substantially higher wages than non-migrants, suggesting migration paid off; this is not causal, as factors apart from age or education may also contribute to observed differences. These gains, however, were predicated on positive labor market outcomes in the destination country. Following the Spanish recession, some migrants chose to return, and although these remained more educated than those who had never left Ecuador, analysis reveals that return migrants had lower levels of human capital than those who chose to stay in Spain.

Return migrants are negatively selected on education with respect to migrants who remain in Spain, but are nevertheless more educated than native-born peers in Ecuador. I pool return migrants identified in full-count Ecuadorian 2010 census microdata (respondents who lived in Spain five years prior but now consider their usual residence to be Ecuador) with Ecuador-born respondents in the 10% iPUMS sample of the Spanish 2011 census (INEC 2026b; INE n.d.[a]; Ruggles et al. 2025a). Returnees are significantly less female (7.7 percentage points less), and appear drawn from the tails of the migrant educational distribution: they are significantly more likely to have a college education (4.4 percentage points, or 67%) and some primary schooling (5.6 percentage points, or 28%), accounting for an age quadratic. Returnees are significantly more male (7.7 percentage points more) and more likely to be married or in a civil union than migrants who remain in Spain (by 20 percentage points, or a 37% increase). In line with labor-market induced return migration, return migrants are 5.5 percentage points (9.7%) more likely to be employed than migrants who remained in Spain. The implications of an inflow of return migrants would depend on how their characteristics compare to the population in Ecuador; from this perspective, return migrant selection in 2010/11 strongly resembles initial emigrant selection for 2007. Table D.6 reveals larger gender differences (returnees are 6.6 percentage points more likely to be male than other native-born Ecuadorians), and higher levels of human capital controlling for an age quadratic, with return migrants possessing secondary schooling at higher rates than their non-migrant Ecuadorian peers (23 percentage points, or 60% more). Despite being more male and possessing higher levels of education, Spanish returnees are 8 percentage points less likely to be employed than non-migrants, and this gap widens once gender, education, experience, urban status and parish fixed effects are taken into account: male (female) re-

nativity, so I exclude respondents whose last residence was abroad.

turn migrants are 12.5 (16) percentage points less likely to work than non-migrants (see Table D.7). Taken together, return migrant selection points to an influx of relatively more educated male workers in need of jobs, which would put downward pressure on domestic wages.

The discussion on return migrants addresses embodied human capital and indicates that, in some cases, migrant or return migrant inflows can stimulate changes to industrial composition and productivity (Hausmann and Nedelkoska 2018). However, the only broader activity in Ecuador with higher employment propensity for male returnees is market services, concentrated within transportation and hospitality. Female returnees are also more likely to work in market services, mostly in hospitality. This may reflect return migrants investing in self-employed jobs with moderate capital constraints, such as owning a vehicle for transporting individuals, or owning property to run a restaurant. However, these are not the most common forms of employment for return migrants. Hospitality is a common migrant industry in Spain, and returnees may have leveraged hospitality-specific human capital obtained during migration episodes for their businesses in Ecuador. However, returnees are less likely to enter construction, the most common migrant industry in Spain, suggesting little scope for the adaptation of construction-specific human capital obtained in Spain to the Ecuadorian context.

3.4 Paying it Backwards: Migrant Remittances

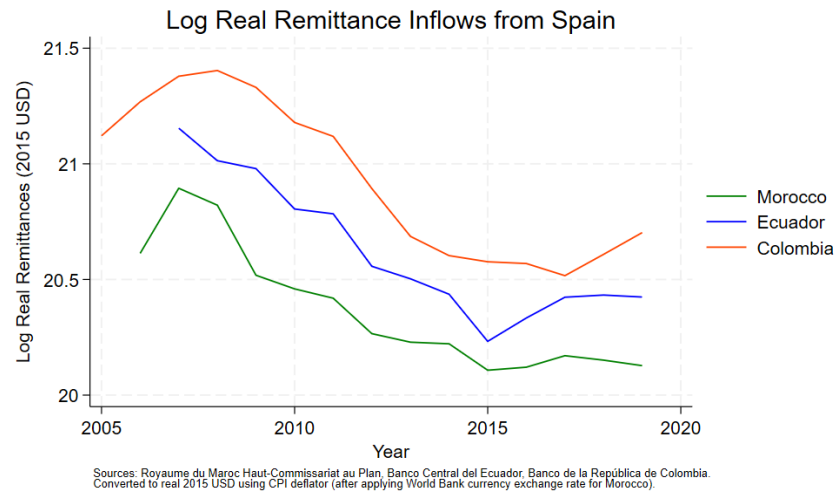
Most working immigrants sent money home in the form of remittances. 67.5% of Ecuadorian migrant workers in Spain remitted funds within the past year (INE 2008b); these remittances averaged 19.8% of monthly wages (see Table D.2). These individual transactions aggregated to substantial financial flows: Spanish remittances to Ecuador amounted to 2.7% of Ecuadorian GDP (Banco Central del Ecuador 2025; WBG 2025a), exceeding official development aid, net FDI inflows, and the export value of bananas, Ecuador’s second largest export good (WBG 2026b; WBG 2026a; World Bank 2026). 5.8% of households received remittances the month prior in December 2007,⁹ and remittances amounted to 43% of monthly income on average for recipients (INEC; 1990–2019). Immigrant labor market integration was positively associated with remittances on the extensive and intensive margins. Employed Ecuadorian migrants were 24.9 percentage points more likely to send remittance than similar unemployed Ecuadorians, while the elasticity of remittances to monthly wages was 0.54 (see Tables D.8 and D.9) (INE 2008b). Similarly impressive are the clear positive correlations between remittances and household wellbeing for Ecuadorian households, as Ecuadorian households receiving remittances in 2007 were materially better off than their peers, more educated, and less likely to be poor (see Table D.10) (INEC; 1990–2019).¹⁰ While not causal, this suggests the large sums sent by migrants had beneficial impacts back home.

During the Global Financial Crisis remittances to Ecuador fell, particularly those from Spain (see Figure 6a). This reduction reflected declines on the extensive and intensive

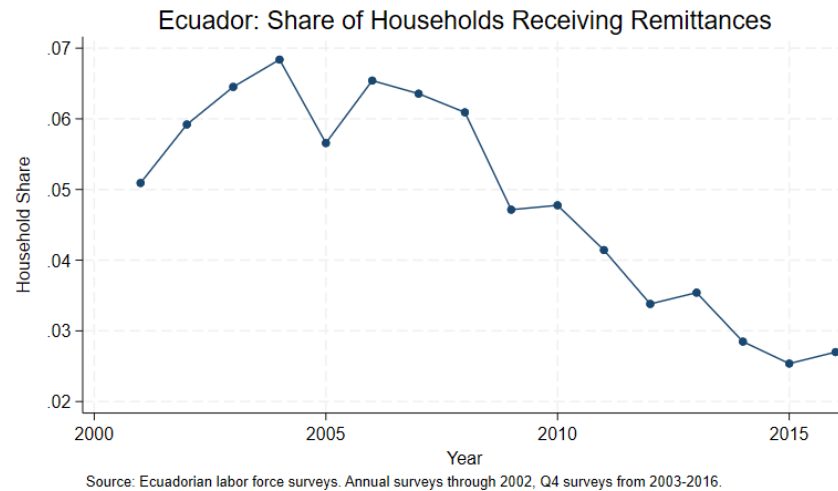
⁹6.99% of households reported receiving remittances within the prior year in the full-count 2010 Ecuadorian census (INEC 2026b).

¹⁰I validate these relationships with 2010 full-count Ecuadorian Census data (INEC 2026b), reported in Table D.11. All relationships that can be tested with the census data retain their statistical significance and valence, while half of those I test have nearly identical coefficients.

margins; both the share of Ecuadorian households receiving remittances (see Figure 6b) and the average funds received per recipient declined over the period (INEC; 1990–2019). Given the relationship between employment and remittance sending, I attribute much of this decline to deteriorating labor market conditions for Ecuadorian emigrants.



(a) Remittances from Spain



(b) ECU: Share of Households Receiving Remittances

Figure 6: Remittance Declines

4 Empirical Strategy

4.1 Theoretical Motivation

My approach is rooted in transnational household decision making with a utility maximization framework, where households decide whether a member should (return) migrate to maximize expected utility based on labor market conditions and migration costs. This parallels Dustmann and Goerlach (2016)'s model of temporary migration, in which migrants decide whether to remain or return depending on realized and expected income across locations. Migration is costly but profitable if expected wages exceed emigration costs (t_d) and the higher cost of living the migrant faces in the destination (C_d). Migration need not be permanent. Migrants decide whether to remain in the destination or return home based on expected wages, return travel costs (t_o), and differences in living costs. All else equal, a negative shock to employment expectations in the destination would reduce emigration and increase return migration.

$$Migr_{i,t} = (E[U_{i,d,t+1}(w_d L_{i,d} - t_d - C_d)] > E[U_{i,o,t+1}(w_o L_{i,o})]) \quad (2)$$

$$Return_{i,t} = (E[U_{i,d,t+1}(w_d L_{i,d} - C_d)] < E[U_{i,o,t+1}(w_o L_{i,o} - t_o)]) \quad (3)$$

A migrant makes a remittance decision based on utility maximization over household consumption in Spain (C_d), their contribution to household consumption in Ecuador (C_o), and their own labor supplied in Spain (L_i). Household consumption in Ecuador (C_o) represents remittances. Consumption in both locations are normal goods, while labor is a disutility. Migrants must cross a subsistence consumption threshold c before sending money to their families back home. Total consumption is constrained by wage labor (wL_i); all else equal, negative shocks to migrant wages or employment would reduce total consumption and remittances.

$$U = f(C_d, C_o, L_i) \quad (4)$$

$$C_d + C_o = wL_i \text{ s.t. } C_d > c \quad (5)$$

Non-migrant household members remaining in Ecuador would operate according to simple labor supply theory. Agents would have a subsistence level of consumption necessary, such that reservation wages change depending on non-wage income, such as remittance income. All else equal, a negative remittance shock will thus lower former remittance recipients' reservation wages and increase the labor supply; this would result in falling real wages if the labor demand curve has a negative slope.¹¹ The basic theory of downward sloping labor demand also implies that an increase in the labor supply from return migration (or changes in net migration) would lower real wages and increase the equilibrium quantity of labor in the market, assuming perfect substitutability between return migrants and non-migrants. Alternatively, if return migrants are complements to non-migrants, as Hausmann

¹¹The remittance impact on labor supply through reservation wages builds on Clemens and McKenzie (2018).

and Nedelkoska (2018) contend, return migration should increase wages and labor supply for complementary non-migrants.¹² The Ecuador migration setting could allow me to test these mental models.

4.2 Estimating Equations

What impact did Ecuadorian emigrant job loss in Spain have on Ecuadorian regions of origin, particularly labor markets and socioeconomic outcomes? Ideally, I would regress changes in socioeconomic outcomes for Ecuadorian regions against emigrant job loss for each region. This has the advantage of accounting for local spillovers and computing net effects for Ecuadorian regions. The independent variable would be the population-adjusted change in employment for Spanish emigrants from region i , $\frac{\Delta Emigr Emp_{i,Spain,t2-t1}}{Pop_{i,t1}}$; this is effectively the change in the local employment rate from emigrant employment. I cannot directly measure this as I do not have time-series data identifying immigrant workers in Spain by nationality, let alone by subnational region of origin.

$$\Delta y_{i,t2-t1} = \beta \frac{\Delta Emigr Emp_{i,Spain,t2-t1}}{Pop_{i,t1}} + \gamma y_{i,t1} + \epsilon_i \quad (6)$$

To estimate the impact of migrant exposure to Spanish downturns on Ecuadorian labor markets, I regress regional changes in wages and employment rates against a migrant shift-share instrument (see Equation 7). The instrument proxies for the share of a region's adult population that would be expected to lose a job in Spain, based on where a region's migrants had settled in Spain and trends in migrant-intensive industries during the Spanish downturn. I compute a weighted average of a Spanish migrant-industry shift-share instrument using local Ecuadorian emigrant distribution across Spain, standardized using emigration rates. This technique resembles the methodology of Caballero, Cadena, and Kovak (2023), except they focus on realized employment changes for conational emigrants, while I instrument for destination labor market conditions using local industry composition and national industry trends (see Appendix B for a robustness check using an instrument more closely approximating their methodology). I estimate subnational emigration $\widehat{Emigr}_{i,j,2007}$ using the Spanish migrant survey and Ecuadorian census microdata as described in Section 2. I compute regional population denominators $Pop_{i,2010}$ from the 2010 Ecuadorian census and use Spanish labor force survey data to compute industry-level employment shares and changes in Spain (INE 2008b; INEC 2026a; INEC 2026b; INE 2024).

$$\Delta y_{i,t2-t1} = \beta \left[\sum_j \frac{\widehat{Emigr}_{i,j,t1}}{Pop_{i,2010}} \sum_l \left(\frac{Emp_{j,l,t1}}{Emp_{j,t1}} \Delta \log (Emp_{-j,l,t2-t1}) \right) \right] + \gamma y_{i,t1} + \epsilon_i \quad (7)$$

I instrument for Spanish local labor market downturns using Spanish labor force survey data. $\frac{Emp_{j,l,t1}}{Emp_{j,t1}}$ records Spanish province j 's employment share in industry (or job type) l

¹²Hausmann and Nedelkoska (2018) argue that, in the post-GFC Albanian case, return migrants from Greece were more entrepreneurial and adopted new technology from abroad, and hence complemented the non-migrant labor force.

in 2007, while shifts come from national log changes in employment, not counting province p , between 2007 and year t , $\Delta \log (Emp_{-j,l,t2-t1})$, restricting to prime-age (25-54) adults, focusing on full-time (30hrs/week+) employment in Q2 labor force surveys (INE 2024). Employment shifts consider full-time, prime-age employees in sectors $l \in L$ irrespective of nativity. I consider employment shares over a set of economic activities L , and I construct candidate instruments using different domains of L . My preferred instrument restricts L to those activities (split by gender) to which Ecuadorian migrants would be most exposed (INE 2008b; INE 2024).¹³ Another version incorporates all ten 1-digit activity-codes identified in the public-use microdata (pooling genders) into L . I can also create singleton sets L , restricting to a single industry or job type l . I create a construction employment instrument, building on Mitze (2019) and Monras (2015) who use construction employment shares to instrument for European and US employment shocks during the Global Financial Crisis, and I also create a temporary-contract instrument.

My instrument contains an implicit first-stage: the underlying Spanish employment shift-share should predict changes in local migrant employment outcomes in Spain. I validate this using Equation 8, which regresses Spanish province-level labor market changes on a domestic Spanish industry employment shift-share instrument:

$$\Delta y_{j,t-2007} = \beta \sum_l^L \left(\frac{Emp_{j,l,2007}}{Emp_{j,2007}} \Delta \log (Emp_{-j,l,t-2007}) \right) + \gamma y_{j,2007} + \epsilon_i \quad (8)$$

I confirm this Spanish first-stage in the following subsection before proceeding to estimate Spanish employment shock impacts on Ecuador in Section 5.

4.3 Spanish First Stage

4.3.1 Employment

I validate the implicit first stage of the migrant shift-share instruments, that a Spanish shift-share instrument predicts changes to migrant employment within Spain. Spanish province-level employment shift-share regressions using Equation 8 demonstrate that the instruments successfully predict changes in local labor market conditions, including for migrants. I investigate several labor-market related outcomes y_j by computing changes to log employment, employment rates, and population for the prime-age (25-54) adult population, and regress these changes on shift-share employment instruments to examine effects over short (2007-09) and medium (2007-11) time horizons. In addition to the preferred instrument which focuses on Ecuadorian migrant intensive gender-industries, and an instrument including all one-digit activities, I examine two simple instruments with singleton sets L focused on construction and temporary jobs, respectively. These are candidates for the shifts within the

¹³Namely: male construction, male trade/hospitality, male agriculture, male simple manufacturing, male complex manufacturing, female other services, female trade/hospitality, female finance/professional services, and female simple manufacturing. Gendered industries based on the 2007 migrant survey and labor force survey. Industry categorization in public-access files for Spanish EPA is coarser than the immigrant survey; these industries were identified by comparing employment distributions of South Americans and Ecuadorians by gender between both datasets.

full Ecuador-focused emigrant employment shift-share instrument described in Equation 7. In the short-run, the simple instruments are most predictive of employment changes (see Table D.12), while over the medium run all instruments are strongly significantly related to changes in all labor market outcomes for Spanish provinces (see Table D.13). Instrumented employment changes in construction or temporary jobs are significantly positively associated with changes in actual jobs over the short and medium run – since these declined over the period, this means regions with greater pre-crisis construction or temporary job employment shares lost more jobs. The negative coefficient on population is somewhat surprising, as it indicates population growth in areas with the largest declines in construction jobs, although the coefficients are roughly half that of log employment. This may be due to greater stability in population trends than employment if construction rates were higher in areas with greater pre-period population growth. These first regressions demonstrate that Spanish shift-share instruments predict general labor market trends; I next demonstrate this includes migrant labor market conditions.

Most importantly, Spanish instruments predict changes to migrant employment in Spain, including for South American migrants. Publicly available Spanish EPA microdata do not identify foreign-born respondents’ exact country of birth, so I examine changes for the foreign-born and South American-born over the short and medium run in Tables 1 and 2. In the short run, the Ecuadorian migrant industries instrument is a more significant predictor of changes to South American log employment than the construction instrument, but only the construction instrument is significantly related to changes in the South American employment rate. However, all candidate instruments are significantly related to changes in migrant and log South American jobs in the medium run, with the construction and migrant industry instruments having the largest coefficients on log employment changes (see Table 2).

These results provide strong evidence that these employment instruments predict local regional labor market downturns within Spain, including for South American migrants, establishing the first step required for these Spanish labor market shocks to reverberate to migrant origin regions abroad.

Table 1: 2007-2009 Province Migrant Employment Changes in Spain

	Log Jobs	Log Migrant Jobs	Log S. Amer Jobs	Emp Rate	Migrant Emp Rate	S. Amer Emp Rate
Construction Bartik	1.529** (0.673)	8.335*** (2.160)	8.565* (4.485)	1.340** (0.508)	2.564*** (0.761)	2.152* (1.096)
Temporary Job Bartik	1.241*** (0.430)	4.445** (1.864)	2.692 (4.502)	1.334*** (0.330)	1.427*** (0.479)	1.266 (0.815)
All Industries Bartik	0.551 (0.614)	4.669*** (1.603)	4.523 (3.510)	0.480 (0.407)	1.012* (0.517)	0.983 (0.945)
Migrant Industries Bartik	0.903 (0.601)	6.280*** (1.447)	7.759** (3.769)	0.654 (0.441)	1.305** (0.494)	1.411 (0.873)
<i>N</i>	52	52	50	52	52	50

Table 2: 2007-2011 Province Migrant Employment Changes in Spain

	Log Jobs	Log Migrant Jobs	Log S. Amer Jobs	Emp Rate	Migrant Emp Rate	S. Amer Emp Rate
Construction Bartik	1.375*** (0.502)	6.839*** (1.661)	11.060*** (2.733)	1.126*** (0.330)	1.619*** (0.438)	2.101** (0.812)
Temporary Job Bartik	1.847*** (0.471)	5.901** (2.602)	7.354* (4.005)	1.926*** (0.366)	2.106*** (0.414)	2.909*** (0.858)
All Industries Bartik	1.162** (0.435)	6.255*** (1.451)	10.173*** (2.490)	0.812*** (0.296)	1.456*** (0.418)	1.787* (0.895)
Migrant Industries Bartik	1.362*** (0.457)	6.952*** (1.466)	11.609*** (2.480)	0.930*** (0.322)	1.598*** (0.395)	2.026** (0.839)
<i>N</i>	52	52	50	52	52	50

Each cell is a coefficient from a unique regression, with its standard error in parentheses. Sample size is the number of Spanish provinces in each column's regressions. Regressions weighted by 2007 prime-age population. Outcome variables are changes in the column variable for prime-age (25-54) populations in Spain for full population, foreign-born, and South American-born. Regressions include 2007 value of the column variable as a control. Heterskedasticity robust (HC2) standard errors. Source: Spanish LFS.

* represents 10% significance, ** 5% significance, *** 1% significance.

4.3.2 Mobility

I test for impacts on Ecuadorian mobility using Spanish population registry moves data from 2006-2019 (INE 2022a) to investigate whether local labor market shocks caused increased return migration to Ecuador, as Section 4.1 would predict. The population registry moves data records internal and external moves involving Spain, and documents individuals country of origin as well as the origin and destination of each move; for moves into or out of the country, this may record the move’s origin or destination country. Özgüzel (2021) cautions that while immigrants face incentives to register their arrivals with local authorities, they have little incentive to record departures. In response, Spain updated its methodology in 2006 strike foreign nationals from population registries if they had not renewed their registration within the past two years (INE 2026b). The subsequent decline in total population registry counts of Ecuadorians in Spain suggests that earlier population numbers were inflated (INE 2023).¹⁴ Additionally, most Ecuadorian “departures” from Spain are cancellations resulting from individuals not confirming their residency, and only 6-42% of annual “departures” for the Ecuadorian-born are identified as return migrations to Ecuador. Departure data is thus inaccurate, since one does not know whether a given cancellation truly represents an out-migration episodes, and even if it did, the destination is unclear and the actual departure year likely inaccurate given the methodology for cancellation. Moves identifying origin or destination countries are more likely accurate, even if such moves, especially departures and return migration episodes, are undercounted.

I regress province-level changes in migration flows and stocks for the Ecuadorian born against Spanish employment shift-share instruments (see Tables 3 and 4). I compute the inverse hyperbolic sine of officially documented return migration, immigration, and net migration between Ecuador and Spain. Changes between 2007-2011 in the inverse hyperbolic sine of net immigration flows of the Ecuadorian-born, restricting to moves identifying Ecuador as the origin or destination country, are significantly related to the 2007-2009 Spanish construction employment instrument in the direction one would expect – greater employment shocks predict larger declines in net immigration. Separate regressions of changes in the inverse hyperbolic sine for both immigration and return migration return coefficients of the expected direction, suggesting reduced immigration and increased return migration; however, these are not significant at the 10% level (p-values of 0.11 and 0.12). I interpret these results as strongly suggestive that migration shocks led to increased Ecuadorian return migration from those exact Spanish regions which experienced greater employment shocks. I also demonstrate declines in official Ecuadorian-born migrant stocks in these regions, but these last results must be interpreted with caution given out-migration underreporting and the methodological change in 2006. I find significant decreases in Ecuadorian-born migrant stocks, but concerning negative pre-trends which could be due to the methodological change that reduced migrant stocks starting in 2006. While the migrant stock regression results could reflect increased return migration in response to local labor shocks, local migration

¹⁴This methodological change triggered a sharp drop in the official Ecuadorian stock, which declined by about 50,000 (over 10%) between 2005 and 2007 (INE 2023). This decline appears artificial; according to official transit statistics from Ecuador, net departures to Spain, while slightly negative in 2005, were positive in 2006, suggesting the Ecuadorian immigrant population actually increased between 2006 and 2007 (INEC 2018).

stocks would reflect changes to net internal migration as well as net international migration. The subsequent Ecuadorian analysis confirms the suggestive finding of increased return migration, as the migrant employment shift-share instrument is significantly associated with increased return migration within those Ecuadorian regions most exposed to Spanish labor market downturns.

Having validated the instrumented Spanish employment shifts, I proceed to run regressions for the main specification to investigate Spanish shock impacts on Ecuadorian regions. I incorporate population weights, weighting by a region’s 2010 prime-age population, and run regressions at both the province and canton level, correcting standard errors to account for small sample size and instrument correlation within provinces. Since my canton-level emigration estimates depend on province-level emigration estimates (see Section 2), the resulting subnational migration matrix assigns cantons within the same province the same relative proportional exposure to Spanish regions across Spain. That is, cantons within the same province have correlated instrument values. To account for this, I cluster standard errors by province when running canton-level shift-share regressions. Given the small sample size for provinces and the small number of clusters, I compute wild bootstrap confidence intervals for the Ecuadorian regressions.

Table 3: 2007-2009 Province Ecuadorian Migration Changes in Spain

	asinh Net Migr Flow from ECU	asinh Return Migr to ECU	asinh Immigr from ECU	log ECU stock
Migrant Industries Bartik	8.77* (3.97)	0.22 (8.12)	6.36* (3.51)	2.33** (0.79)
Construction Bartik	4.13 (5.04)	-7.86 (10.05)	2.92 (4.38)	2.07** (1.00)
Temporary Job Bartik	0.13 (3.13)	6.05 (6.16)	0.35 (2.71)	1.02 (0.65)
All Industries Bartik	8.28* (3.77)	-0.45 (7.66)	5.88* (3.34)	2.32*** (0.73)
<i>N</i>	49	50	50	52
Province Avg Value (2007)	6.955 (1.782)	3.341 (2.132)	6.968 (1.791)	9.040 (1.948)
Province Avg Value (2009)	6.050 (1.748)	4.367 (1.924)	6.230 (1.746)	9.161 (1.919)

Table 4: 2007-2011 Province Ecuadorian Migration Changes in Spain

	asinh Net Migr Flow from ECU	asinh Return Migr to ECU	asinh Immigr from ECU	log ECU stock
Migrant Industries Bartik	158.30** (43.96)	1.65 (9.27)	14.12** (5.18)	4.24*** (1.20)
Construction Bartik	144.13** (56.57)	-13.64 (11.38)	10.68 (6.59)	4.24*** (1.53)
Temporary Job Bartik	41.64 (36.84)	1.88 (7.10)	2.13 (4.16)	1.43* (1.03)
All Industries Bartik	142.21** (42.37)	0.31 (8.75)	11.47** (5.03)	4.31*** (1.09)
<i>N</i>	49	50	50	52
Province Avg Value (2007)	6.955 (1.782)	3.341 (2.132)	6.968 (1.791)	9.040 (1.948)
Province Avg Value (2011)	1.460 (4.146)	5.230 (1.988)	5.400 (1.736)	9.190 (1.874)

Each cell is a coefficient from a unique regression, with its standard error in parentheses. Sample size is the number of Spanish provinces in the final regression of each column. Regressions weighted by 2007 prime-age population. Outcome variables are changes in the column variable in Spain. Regressions include 2007 value of the column variable as a control. Wild bootstrapped, heteroskedasticity robust (HC2) standard errors. Source: Spanish LFS, population registry, population registry moves.

* represents 10% significance, ** 5% significance, *** 1% significance.

5 Results

The question this paper attempts to answer is how Ecuadorian emigrants losing jobs in Spain impacted labor markets and socioeconomic outcomes within Ecuador. In the prior section I designed an emigrant employment shift-share instrument to answer this question, and demonstrated that its implicit first stage holds, as instrumented Spanish shifts do predict migrant job loss within Spain. Why should emigrant job loss matter for a migrant sending region? First, as migrant job loss causes income reduction in Ecuador by leading remittances to fall, and second by increasing local labor supply by incentivizing return migration. As described in Section 3.4, remittance sending and amounts are positively associated with migrant employment and wages, respectively; this suggests reduced migrant employment and income should lead to less remittances sent home. Remittances are a significant source of income for recipient households (recall from Section 3.4 that remittances on average amounted to 43% of monthly income for recipients (INEC; 1990–2019)); reduced remittances should impact local household labor decisions by reducing reservation wages and thus increasing employment rates. It would also immediately reduce recipients’ budget constraints, reducing consumption. For migrants, job loss and declining employment prospects would reduce utility in Spain, leading some to return migrate. Return migration would impact a sending region to the extent it changes labor supply, human capital, and labor demand. Spanish employment shocks would propagate to sending regions by changing international financial and human flows, with each mechanism having potentially distinct implications.

Declining remittances and increased return migration may have overlapping or counter-vailing impacts on labor markets in sending region. Declining remittances would reduce non-wage incomes for transnational households, lowering reservation wages and prompting households to increase their domestic labor supply, shifting out the labor supply curve. At the same time, falling household income would reduce local labor demand. Taken together, wages would fall, with ambiguous net effects on overall employment. Simultaneous shock transmission through return migration could have ambiguous effects of its own, depending on whether return migrants are labor market substitutes or complements to non-migrants, for instance by implementing new technologies that trigger agglomeration economies. If return migrants are perfect substitutes in production, the increase in the labor supply would shift the labor supply curve out, building on the remittance shock effect. Even as perfect substitutes, return migrants could also increase local labor demand, creating an ambiguous total effect on wages, but a positive total effect on employment.

However, if return migrants operate as complements to domestic labor due to segmented labor markets, for example due to entrepreneurship or the adoption of new technologies as in Hausmann and Nedelkoska (2018), return migration would increase labor demand for non-migrant workers, resulting in higher wages and employment for non-migrants.

I first confirm that the remittance and return migration mechanisms are at play by regressing local changes to remittance inflows and return migration rates against the emigrant employment shift-share instrument. To compute net effects on local labor markets, I then regress local changes in employment rates and wages against these instruments, running separate regressions by gender, age, and education cohorts. My results confirm both mechanisms are active, as total remittance inflows decline while return migration increases. Labor

market results indicate positive employment rate responses, in line with the reduced income channel, as employment among older cohorts and female cohorts increases. I find evidence of wage declines, particularly for less educated cohorts, suggesting these employment increases created downward pressure on wages.

5.1 Mechanisms

First, I confirm that the Spanish employment shock operates through both remittance and return migration mechanisms, using migrant shift-share instrument regressions to reveal the reverberations of Spanish labor market shocks to Ecuador. Regressions follow Equation 7, reproduced below, and are conducted at historic province and canton-levels in Ecuador, weighting observations by 2010 prime-age population and computing wild-bootstrapped standard errors (clustered by province for canton regressions).¹⁵ Δy represent changes in multi-year averages, rates or medians.¹⁶ For ease of interpretation, I reverse the sign of the migrant instruments so a positive (negative) regression coefficient indicates exposure to Spanish labor market downturns is associated with a relative increase (decrease) in an outcome for an Ecuadorian region. One would expect a decline in remittances and increase in return migration due to reduced migrant employment, and my analyses confirm these both took place.

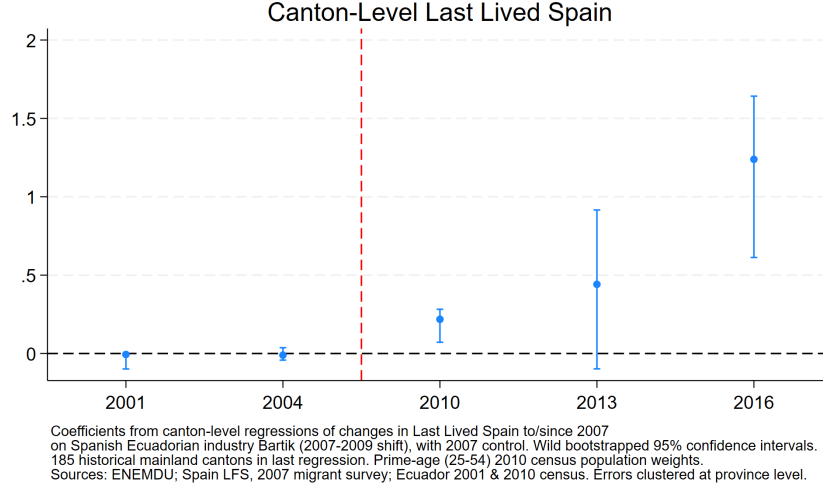
I find that the primary mechanisms for shock diffusion, remittances and return migration, respond to emigrant employment shocks as expected at both the province and canton levels. For remittances, $\log(\text{rem})_i$, I investigate changes in log multiyear average total real remittance inflows at province and canton levels, as recorded by Banco Central del Ecuador (2025). The Ecuadorian Central Bank only reports aggregate remittance flows at the province and canton level, so I cannot isolate Spanish remittances. I proxy for return migration rmigr_i from Spain using the ENEMDU as the proportion of the prime-age population that report Spain as their prior location of residence.

At the canton level, I find statistically significant relative increases in return migration rates from Spain and decreases in total log real remittance inflows in the short and long run (see Figure 7). While I find significant decreases in total remittance inflows, I find no significant relative decline in local household remittance rates from ENEMDU surveys.¹⁷ I find similar results at the province level, with rising return migration and falling log remittances (see Figure C.7). For robustness, I investigate alternate emigrant shift-share instruments in Table B.1, and generally find similar results. This analysis indicates that migrant employment shocks propagated from Spain to Ecuador through increased return migration and reduced remittances. I next test for implications in Ecuadorian labor markets.

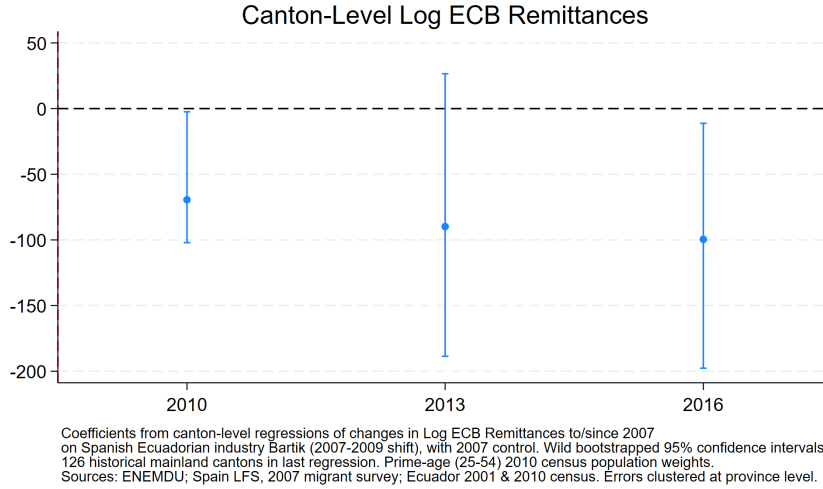
¹⁵Ecuador provinces are subdivided into cantons, with the smallest cantons having populations in the low thousands. Not all cantons are consistently covered in ENEMDU or ECB data, and my current geographic standardization (given changes in cantons over time) may leave out a small number of cantons.

¹⁶ Years are pooled as follows: 2000-2001, 2003-2005, 2006-2008, 2009-2011, 2012-2014, 2015-2017, 2018-2019, resembling a strategy used with ENEMDU data in Ontaneda Jiménez et al. (2024). I indicate series with the midpoint year for 3-year averages or last year for 2-year averages.

¹⁷Both ECB and ENEMDU time-series have incomplete canton coverage; ENEMDU has greater coverage.



(a) Canton Share Last in Spain



(b) Canton Log Remittance Inflow

Figure 7: Canton Mechanism Regressions

5.2 Labor Market Impacts

5.2.1 Main Results

To examine the effects of the Spanish employment shock on local labor markets in Ecuador, I investigate regional changes in employment rates and wages. Regression outcomes are province and canton-level changes within Ecuador for full-time (30+ hrs/week) employment rates, $emprate_{i,c}$, and log real hourly wages for full-time workers, $\log(w)_{i,c}$, focusing on prime-age (25-54) respondents. To investigate heterogeneous effects, I run distinct regressions for various gender, education, and age cohorts c . Employment rates are multi-year weighted means, while log hourly wages are multi-year weighted medians, computed for historic mainland Ecuadorian provinces and cantons from pooled ENEMDU survey data, applying survey

weights.¹⁸ Self-employment rates are the share of a cohort reporting that they are self-employed and working full-time (30+ hrs/week). I first investigate canton changes for all prime-age workers.

Regressions of labor market changes against emigrant employment shift-share instruments reveal significant increases in full-time employment rates and self-employment rates in the short run (2007-2010) (see Table 7). Self-employment rates continue to grow in the medium run (2007-2013) despite full-time employment rates ceasing to show significant differentials. Nationally, self-employment appears counter-cyclical in Ecuador (Ontaneda Jiménez et al. 2024), so the increase in self-employment could reflect local economic distress from the emigrant employment shock. In addition, I find negative but insignificant impacts on wages for all prime-age (25-54) workers in the short and medium run. Rising employment rates are in line with former remittance recipients increasing work in response to decreased reservation wages due to reduced remittance inflows. The null result on wages suggests no overall wage impacts, but a heterogeneity analysis by education reveals downward wage pressure for less educated workers in Table 8. For education I distinguish between those with and without some secondary education (i.e. high school). I run separate regressions by gender and education to examine heterogeneous effects across these dimensions.

Gender and education analyses reveal gendered employment effects and education-specific wage effects. Full-time employment rate increases are concentrated among women, and self-employment rate increases among men (see Table 7). This suggests that women entered the labor market, while employed men shifted into self-employment in response to the shock. Employment changes were education-neutral, however. In contrast, wage changes were gender-neutral and education-specific, with significant negative coefficients for less educated workers contrasting with insignificant positive coefficients for educated workers in Table 8. This suggests that increased female employment exerted downward wage pressure for less educated workers. To further investigate heterogeneity, I run a more detailed cohort analysis accounting for age.

Examining gendered age cohorts – individuals born within the same decade – reveals that employment rate responses were strongest for older cohorts of both genders. Men and women aged 58-67 in 2007 (1940s birth cohort) have strongly significant increases in their full-time employment rates between 2007 and 2010 (see Table D.14). Men aged 48-59 (1950s cohort) and women aged 28-39 (1970s cohort) have mildly significantly positive increases in full-time employment. The largest coefficients are for women; the coefficient of 11.92 for women aged 58-67 (1940s birth cohort) implies that a 1 percentage point decrease in the canton’s employment rate from emigrant construction job loss would result in a 11.92 percentage point increase in the full-time employment rate for women aged 58-67 in the three years following the recession’s onset. Since average cohort employment rates were declining for the oldest cohorts (from 26.3 to 22.7 percent for women, and 68.0 to 60.0 percent for men), the relative increases may be due to deferment of retirement rather than entry into the labor force. Female employment is countercyclical in Ecuador (Ontaneda Jiménez et al. 2024), so these relative increases in female full-time employment rates may reflect local economic distress due to reduced remittances.

¹⁸Given multiyear averages, this is the difference between the 2006-8 and 2009-11 averages. See Footnote 16 for details on year aggregation.

Running wage change regressions by gender and gendered age cohorts for less educated workers reveals that these wage declines were concentrated among women, although young men were also affected (see Table D.15. All coefficients for these cohorts of workers without secondary school education are negative. Pooling by gender, wage changes for female workers without secondary schooling are significantly negative. The coefficient of -22.78 from the construction instrument for median log real hourly wages suggests that a 1 percentage point decrease in the canton’s employment rate from emigrant construction job loss led to a 0.2 log point decrease in hourly wages for women without secondary education, roughly a 20 percent decrease in real hourly wages. Since average log median hourly wages rose for this demographic (from 1.2 to 1.5), the results could reflect wage stagnation rather than absolute declines in wage rates. Investigating different age cohorts, I find significant negative coefficients for middle aged women aged 48-57 in 2007 (1950s cohort), and mildly significant negative coefficients for younger female cohorts aged 18-27 and 28-37 (1980s and 1970s birth cohorts). While most less educated male cohorts have insignificantly negative coefficients, those aged 28-37 in 2007 have significantly negative coefficients for all three included instruments. The construction instrument’s coefficient of -12.57 suggests a 1 percentage point decrease in the canton’s employment rate from emigrant construction job loss led to a nearly 0.13 log point decrease in hourly wages for men, roughly a 13% decrease in hourly wages. Falling wages might arise from labor competition from older workers, whose employment rates increased. Alternatively, this could be driven by competition from return migrants, as in Edo (2020), who found that French regions with greater exposure to French Algerian immigration experienced downward wage pressure for less educated non-migrants.

I test for impacts on industry employment. Similar to prior rates, these are multi-year means computed from ENEMDU data, restricting to full-time workers so each fraction is a subset of the full-time employment rate. Thus, $constr_{i,c,t}$ is the share of cohort c in canton i working full-time in construction in year t . I run these regressions for prime-age (25-54) respondents in Table 6, and find a mildly significant short-run decline in the employment share in construction, but insignificant changes for other industries. Tables D.16 and D.17 represent industry employment rate change regressions by gender. This reveals the overall construction decline is due to declining male employment in construction, which is significant at the 5% level. The coefficient of -1.54 for the construction bartik indicates a 1 percentage point decrease in a canton’s employment rate from emigrant construction job loss would have led to a 1.57 percentage point decline in the construction employment rate within the same canton. This decline in construction employment likely reflects a drop in demand for construction from emigrants, as remittances contribute to housing demand.¹⁹ Ironically, the loss of construction employment in Spain thus led to an immediate drop in construction employment in Ecuador as well.

5.2.2 Robustness Checks

I run additional regressions incorporating various controls for robustness. The main results – increased employment and self-employment due to emigrant job loss – could be spurious if

¹⁹Jokisch and Kyle (2008) note that Ecuadorian emigrants often funded housing construction in Ecuador; Table D.10 indicates homes for remittance-recipients are larger and have better amenities, suggesting remittance-funded construction projects.

omitted factors, which happen to be correlated with the emigrant employment instrument, actually drive these changes. For instance, government expenditures or policies with regionally heterogeneous effects that correlate with Spanish emigrant job loss or emigration to Spain could be responsible for these changes. In this case, the regression results would overstate the effect of emigrant job loss. Omitted variable bias could also lead the effects of emigrant job loss to be underestimated. One candidate for such an underestimation is migration to other destinations, as I only examine emigration to Spain. Ecuadorian emigrant stocks were nearly evenly split between Spain and the United States at the eve of the Global Financial Crisis (UN 2020), but subnational emigration rates to both countries are uncorrelated.²⁰ The United States experienced job loss during the Great Recession, so Ecuadorian cantons with emigration links to the USA were exposed to emigrant job loss simultaneous to cantons with emigration links to Spain. However, since the instrument only incorporates emigrant job loss from Spain, cantons exposed to emigrant job loss in the USA would be considered as “untreated,” which would bias coefficients downwards. Omitted variable bias may thus either overestimate or underestimate my coefficients.

To address these concerns, I re-run several main regressions with various sets of controls. I add employment sector, public employment growth, and human capital controls from ENEMDU, regional sectoral production statistics from the Ecuadorian Central Bank, and emigration data from Ecuadorian censuses. I first control for employment shares by industry according to ENEMDU – the share of the prime age population working full-time in five large standardized industrial groupings (agriculture, manufacturing, market services, non-market services, and construction) in 2007 (INEC; 1990–2019). Next, I test for local industry composition through sectoral shares of total canton production, using Ecuadorian Central Bank canton-level data on production by sector (ECB 2026).²¹ To control for government spending, the third suite of controls adds the share of the prime age population working full-time in the public sector in 2007, and its change between 2007–2010, according to ENEMDU. The fourth suite of controls focuses on log growth in sector production between 2007–2010, restricting to more common sectors.²² The fifth set of controls examines initial human capital differences, taking the share of the prime age population whose education ends in primary education, secondary education, and superior education (no education would be the omitted category). The last control includes the prime-age population emigration rate to the United States, derived from the 2001 and 2010 Ecuadorian census (this is the proportion that a canton’s prime-age population would increase if all prime-age emigrants to the United States between 1995–2007 returned in 2010). I re-examine regressions for three significant labor market outcomes – changes in prime-age full-time employment rates, self-employment rates, and male construction employment rates – with each of these six suites of controls.

²⁰I compute the prime-age population-weighted correlation of prime-age emigration rates to Spain and the US using Ecuadorian census emigration data (I proxy the emigration rate by dividing canton prime-age emigration flows from 1995–2007 by canton prime-age population stocks in 2010) is -0.02 (INEC 2026a; INEC 2026b).

²¹Sectors include: agriculture, commerce, construction, education, finance, health, hospitality, manufacturing, mining, professional services, public administration, transportation and communications, and other services; I omit utilities to avoid multicollinearity. Production is in USD, which I convert to real 2014 USD using Ecuadorian CPI.

²²These are: agriculture, commerce, construction, education, manufacturing, professional services, public administration, transportation and communications, other services, and utilities.

The relationship between the Spanish emigrant employment instrument and full-time employment shares withstands the battery of controls, but the other outcomes, in particular the male construction employment rate, are sensitive various sets of controls. The full-time employment rate coefficient remains statistically significant and retains its positive sign under each suite of controls, although it becomes only marginally significant at the 10% level under public employment controls. The coefficient for self-employment remains statistically significant after controlling for US emigration, human capital, and initial sectoral GDP shares; it close to doubles in absolute value with both latter sets of controls, but becomes only marginally significant in the education shares specification. However, significance disappears with public employment, sector growth controls, and industry employment shares, and even reverses sign with the industry employment shares. Each set of controls reduces the statistical significance of the coefficient on the male construction employment rate, and the coefficient only retains significance with the public employment and US emigration rate controls. While these robustness checks appear to confirm that migrant job loss increased full-time employment rates, they caution that observed impacts on domestic construction and self-employment rates may be driven by other factors.

Table 5: 2007-2010 Canton Labor Market Changes, Ecuador

	Emp Rate	Log Hrly Wage	Self-Emp
Construction Bartik	2.92*** (1.57)	-1.87 (3.60)	1.12* (0.60)
Migrant Industries Bartik	2.02*** (1.09)	-1.28 (2.51)	0.81* (0.43)
Emigration Rate	0.16*** (0.08)	-0.09 (0.19)	0.05* (0.03)
<i>N</i>	199	199	199
Canton Avg Value (2007)	0.638 (0.096)	1.745 (0.273)	0.212 (0.054)
Canton Avg Value (2010)	0.634 (0.087)	1.934 (0.215)	0.227 (0.057)

Table 6: 2007-2010 Canton Industry Employment Changes, Ecuador

	Ag	MFG	Mkt Svcs	Non-Mkt Svcs	Constr
Construction Bartik	0.64 (1.69)	-0.17 (0.48)	0.51 (0.74)	1.89 (0.97)	-0.51 (0.37)
Migrant Industries Bartik	0.44 (1.18)	-0.12 (0.34)	0.36 (0.51)	1.32 (0.68)	-0.36 (0.26)
Emigration Rate	0.03 (0.08)	-0.01 (0.03)	0.03 (0.04)	0.10 (0.05)	-0.02 (0.02)
<i>N</i>	199	199	199	199	199
Canton Avg Value (2007)	0.138 (0.144)	0.076 (0.046)	0.236 (0.095)	0.130 (0.055)	0.048 (0.020)
Canton Avg Value (2010)	0.137 (0.141)	0.073 (0.044)	0.238 (0.100)	0.132 (0.051)	0.045 (0.020)

Each cell is a coefficient from a unique regression, with standard error in parentheses. Sample size is the number of Ecuador cantons in final regression of each column. Regressions are weighted by 2010 prime-age (25-54) canton population. Dependent variables are changes in column title between 2007 (2006-8) and 2010 (2009-11) averages in Ecuador for prime-age (25-54) respondents; industries are shares in full-time employment in specified industry grouping. Regressions include 2007 (2006-8) values as controls. Wild bootstrap standard errors and confidence intervals, clustered by province. Shift-share instruments calculated based on 2007-2009 employment changes in Spain.

* represents 10% significance, ** 5% significance, *** 1% significance.

Table 7: 2007-2010 Canton Labor Market Changes by Gender, Ecuador

	Female Emp Rate	Male Emp Rate	Female Log Hrly Wage	Male Log Hrly Wage	Female Self-Emp	Male Self-Emp
Construction Bartik	5.30** (2.58)	-0.82 (0.93)	-10.14 (4.44)	-3.16 (3.51)	0.97 (1.24)	2.50** (1.12)
Migrant Industries Bartik	3.67** (1.78)	-0.59 (0.65)	-7.07 (3.09)	-2.18 (2.45)	0.72 (0.88)	1.76** (0.77)
Emigration Rate	0.28** (0.13)	-0.03 (0.05)	-0.51 (0.24)	-0.15 (0.19)	0.05 (0.06)	0.11* (0.06)
<i>N</i>	199	199	194	199	199	199
Canton Avg Value (2007)	0.443 (0.146)	0.849 (0.077)	1.639 (0.310)	1.793 (0.284)	0.181 (0.054)	0.246 (0.089)
Canton Avg Value (2010)	0.444 (0.139)	0.842 (0.064)	1.885 (0.272)	1.968 (0.229)	0.184 (0.058)	0.273 (0.086)

Table 8: 2007-2010 Canton Labor Market Changes by Education, Ecuador

	Non-HS Emp Rate	HS+ Emp Rate	Non-HS Log Hrly Wage	HS+ Log Hrly Wage	Non-HS Self-Emp	HS+ Self-Emp
Construction Bartik	2.34* (1.78)	5.17* (2.94)	-7.08* (2.78)	10.32 (5.39)	0.51 (1.85)	1.37 (1.76)
Migrant Industries Bartik	1.60* (1.24)	3.64* (2.05)	-4.92* (1.92)	7.28 (3.77)	0.37 (1.32)	0.97 (1.23)
Emigration Rate	0.13** (0.09)	0.28* (0.15)	-0.35 (0.15)	0.56 (0.29)	0.02 (0.09)	0.06 (0.09)
<i>N</i>	199	199	199	196	199	195
Canton Avg Value (2007)	0.591 (0.100)	0.672 (0.088)	1.478 (0.198)	1.958 (0.270)	0.243 (0.073)	0.194 (0.069)
Canton Avg Value (2010)	0.591 (0.098)	0.663 (0.093)	1.695 (0.191)	2.119 (0.208)	0.264 (0.075)	0.204 (0.058)

Each cell is a coefficient from a unique regression, with standard error in parentheses. Sample size is the number of Ecuador cantons in final regression of each column. Regressions are weighted by 2010 prime-age (25-54) canton population. Dependent variables are changes in column title between 2007 (2006-8) and 2010 (2009-11) averages in Ecuador for prime-age (25-54) respondents in designated demographic group.

Regressions include 2007 (2006-8) values as controls. Wild bootstrap standard errors and confidence intervals, clustered by province. Shift-share instruments calculated based on 2007-2009 employment changes in Spain.

* represents 10% significance, ** 5% significance, *** 1% significance.

Table 9: 2007-2010 Canton Full-Time Employment Rate Changes, Ecuador

Migrant Emp Bartik (2007-9)	2.021*** [0.46 , 6.65]	2.642** [0.94 , 6.33]	3.003** [0.93 , 7.21]	1.638* [-0.14 , 6.65]	2.374** [0.78 , 7.55]	2.344** [0.38 , 6.14]	1.997*** [0.44 , 6.72]
Full-time Emp Rate (2007)	-0.236*** [-0.30 , -0.13]	-0.208 [-1.24 , 0.32]	-0.330*** [-0.56 , -0.14]	-0.250*** [-0.35 , -0.11]	-0.257*** [-0.40 , -0.09]	-0.314*** [-0.45 , -0.15]	-0.232*** [-0.30 , -0.12]
Observations	199	199	198	199	193	199	199
Adjusted R2	0.190	0.250	0.295	0.191	0.192	0.259	0.188
Industry Emp Shares		X					
Sector GDP Shares			X				
Public Employment				X			
Sector log GDP growth					X		
Education Shares						X	
USA Emigration Rate							X

Wild bootstrap 95% confidence intervals in brackets, clustered by province.

* p<0.10, ** p<0.05, *** p<0.01

Table 10: 2007-2010 Canton Self Employment Rate Changes, Ecuador

Migrant Emp Bartik (2007-9)	0.812** [0.02 , 2.78]	-0.156 [-3.63 , 3.57]	1.988** [0.45 , 6.18]	1.547 [-0.68 , 4.35]	0.702 [-0.60 , 2.42]	1.494* [-0.10 , 4.06]	0.841** [0.19 , 2.57]
Self-Emp Rate (2007)	-0.325** [-0.54 , -0.09]	-0.432** [-0.71 , -0.13]	-0.423*** [-0.55 , -0.33]	-0.321*** [-0.53 , -0.10]	-0.359*** [-0.54 , -0.18]	-0.379*** [-0.51 , -0.26]	-0.335** [-0.57 , -0.09]
Observations	199	199	198	199	193	199	199
Adjusted R2	0.131	0.182	0.225	0.218	0.138	0.173	0.129
Industry Emp Shares		X					
Sector GDP Shares			X				
Public Employment				X			
Sector log GDP growth					X		
Education Shares						X	
USA Emigration Rate							X

Wild bootstrap 95% confidence intervals in brackets, clustered by province.

* p<0.10, ** p<0.05, *** p<0.01

Table 11: 2007-2010 Canton Male Construction Employment Rate Changes, Ecuador

Migrant Emp Bartik (2007-9)	-1.072**	-1.137	-1.046	-1.131*	-0.749	-1.093	-1.014*
	[-3.36 , -0.13]	[-5.66 , 0.28]	[-4.17 , 0.55]	[-3.46 , 0.28]	[-2.85 , 0.79]	[-4.79 , 0.54]	[-3.21 , 0.13]
Male Constr. Rate (2007)	-0.330***	0.389	-0.475***	-0.334***	-0.369***	-0.372***	-0.358***
	[-0.48 , -0.19]	[-0.37 , 1.39]	[-0.78 , -0.14]	[-0.48 , -0.19]	[-0.61 , -0.14]	[-0.52 , -0.21]	[-0.58 , -0.20]
Observations	199	199	198	199	193	199	199
Adjusted R2	0.193	0.247	0.236	0.185	0.210	0.205	0.207
Industry Emp Shares		X					
Sector GDP Shares			X				
Public Employment				X			
Sector log GDP growth					X		
Education Shares						X	
USA Emigration Rate							X

Wild bootstrap 95% confidence intervals in brackets, clustered by province.

* p<0.10, ** p<0.05, *** p<0.01

6 Conclusion

I find evidence that exposure to Spanish economic downturns through migration networks led to declines in remittance inflows and increased return migration to Ecuador, with impacts for local Ecuadorian labor markets. At the canton level, employment rates and self-employment rates increased for the prime-age population, and a cohort analysis reveals that labor force attachment particularly increased for women and older men. I interpret this as suggestive evidence that households in Ecuador increased their supply of labor in response to compensate for income shortfalls associated with remittance shocks. This collective response may have triggered market forces dampening the efficacy of this strategy. While overall wages remained unchanged, hourly wages for less educated female full-time workers fell, as did those of less educated male workers mostly in their thirties. Impacted regions witnessed relative increases in self-employment; increased female labor force participation and self-employment are both countercyclical in the Ecuadorian context, further suggesting that emigrant job loss produced financial strain for Ecuadorian households. Examining local employment across industries, I find a significant drop in male full-time employment in the construction sector in cantons most exposed to the Spanish construction crash. This suggests that migrant networks can transmit a sectoral employment shock from abroad to the same sector at home. Results for full-time employment rates are robust to controls, although self-employment and construction results are sensitive to employment share controls. Future work would trace the dynamics of employment and wage responses in more detail, particularly for sub-group analysis. In conclusion, I expand on the migration literature by providing evidence of shock propagation of migrant job loss through migrant networks even in a migration corridor spanning significant geographic distance.

A Geographic Crosswalk

I create a canton code crosswalk to geographically standardize regional Ecuadorian data given changes in administrative divisions around the period of study. Cantons have four-digit codes, with the first two digits corresponding to a canton’s province code. I create a canton crosswalk based on 2001 Ecuadorian census microdata geographically standardized by INEC (INEC 2026a), and assign modern canton codes to historical codes based on population distribution (new cantons are assigned to the old canton containing the plurality of its 2001 population; this is over 50% of the population in all but one case). I supplement this standardization to match the Spanish immigrant survey. The 2007 Spanish immigrant survey and codebook (INE 2008b; INE 2008a) omit an Ecuadorian province, Orellana, which split off from the province of Napo in 1998 (*Provincia de Orellana* 2026). I assign cantons in Orellana their codes from before the province split from Napo, consulting administrative boundary classifications from INEC (INEC 2009). I also reclassify a canton of disputed province in 2001 (La Concordia) to its 2010 canton code, which I confirm with geographically standardized 2010 Ecuadorian census microdata (INEC 2026b). The resulting crosswalk standardizes codes to 217 historic cantons across 21 historic provinces (214 cantons and 20 provinces excluding Galapagos).

B Alternative Instruments

My preferred migrant employment shift-share instrument considers both subnational destinations as well as national changes to employment across gendered industries. One may be concerned that the shifts, the predicted employment changes for migrants, would be inaccurate if native and migrant employment moves differently, or to the extent that industry codes include sub-industries that are economically irrelevant for migrant employment. Equation 9 is an alternate instrument focused solely on immigrant employment, adapting Caballero, Cadena, and Kovak (2023)’s migrant employment instrument to the Ecuadorian-Spanish context. In their paper, Caballero, Cadena, and Kovak (2023) use observed changes in total co-national migrant employment for Mexican migrants in the United States as the shifts in their instrument, taking migrant employment changes as given without instrumenting for changes to destination migrant employment. One advantage of this approach is that it focuses on realized employment changes; this sidesteps concerns about including irrelevant sub-industries or workers in an instrument. Nevertheless, there are at least three disadvantages, two conceptual and the other a practical issue that arises when applying the strategy to the Spanish context. First, the instrument takes destination-market employment changes as given, which could present a causality concern if observed changes in migrant employment are due to conditions in the origin country, not the destination country – for example, improving conditions in the origin could trigger return migration and mechanically reduce migrant employment levels. Relatedly, changes in migrant employment can reflect changing migrant stocks in addition to changing migrant employment rates; a 10% decline in migrant employment could be due to 10% of migrants losing their jobs, 10% of migrants leaving the region, or a mix of job loss and out-migration. Finally, data coarseness in the Spanish context introduces measurement error. Publicly available Spanish labor force survey data does not

identify immigrant birth country, only broad region of origin. I can only examine changes in South American migrant employment, not Ecuadorian migrant employment, using this data. Changes to South American employment could reflect nationality-specific employment and migration patterns for non-Ecuadorians, introducing measurement error and biasing the coefficient downwards. In any case, the instrument is as follows:

$$\Delta y_{i,t2-t1} = \beta \left[\sum_j \frac{\widehat{Emigr_{i,j,2007}}}{Pop_{i,2010}} \Delta \log(S.AmerEmp_j) \right] + \gamma y_{i,t1} + \epsilon_i \quad (9)$$

Another view is that pre-crisis emigrant location within Spain is irrelevant, either because migrants face an integrated national Spanish labor market or because of high migrant mobility. Existing research confirms greater mobility for migrants than natives in response to labor shocks in European and Spanish contexts (Mitze 2019; Basso, D’Amuri, and Peri 2019; Özgüzel 2021). In this framework, all that matters for Ecuadorian exposure to Spanish employment shocks is total regional emigration rate to Spain, not subnational destination emigration rates. Equation 10 uses the 2007 local emigration rate to Spain to instrument for exposure to Spanish employment shocks:

$$\Delta y_{i,t2-t1} = \beta \frac{\widehat{Emigr_{i,Spain,2007}}}{Pop_{i,2010}} + \gamma y_{i,t1} + \epsilon_i \quad (10)$$

I compare canton-level mechanism regressions, adding the above instruments to the main instruments discussed in the paper. I find that the South American employment instrument is statistically insignificantly related to the primary mechanisms, but that the other candidate instruments are similarly significant.

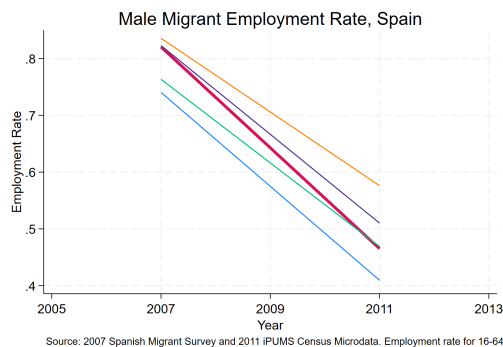
Table B.1: 2007-2010 Canton Mechanism Changes, Ecuador

	Last in Spain	Log Remittances
Construction Bartik	0.31** (0.05)	-98.66* (17.78)
Migrant Industries Bartik	0.22*** (0.03)	-69.39** (12.46)
Emigration Rate	0.02** (0.00)	-4.86* (0.99)
All Industries Bartik	0.19*** (0.03)	-59.55* (11.07)
Temporary Job Bartik	0.23** (0.04)	-75.80* (13.77)
S. American Shift-Share	0.09 (0.08)	-36.34 (40.32)
<i>N</i>	199	128
Canton Avg Value (2007)	0.001 (0.001)	17.852 (2.894)
Canton Avg Value (2010)	0.001 (0.003)	17.691 (2.761)

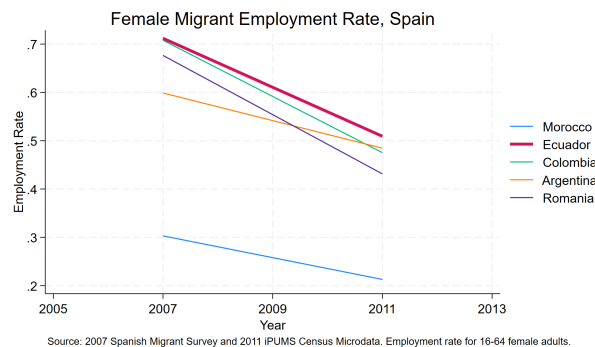
Each cell is a coefficient from a unique regression, with standard error in parentheses. Sample size is the number of Ecuador cantons in each column's final regression. Regressions are weighted by 2010 prime-age (25-54) canton population. Dependent variables are changes in prime-age share having last lived in Spain, and change in log real remittance inflow between 2007 and 2010. Regressions include 2007 values as controls. Wild bootstrap standard errors and confidence intervals. Shift-share instruments calculated based on 2007-2009 employment changes in Spain.

* represents 10% significance, ** 5% significance, *** 1% significance.

C Supplementary Figures

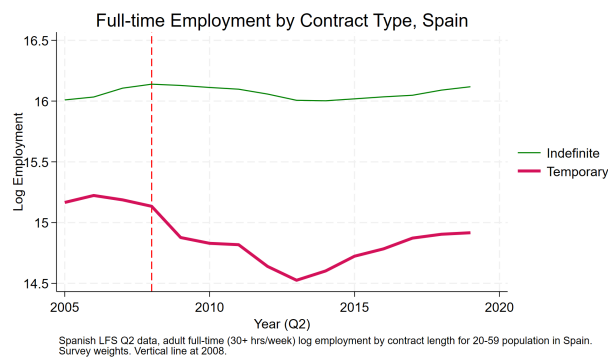


(a) Male Employment Rates

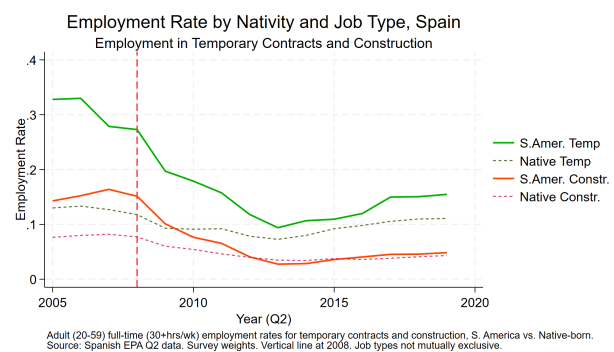


(b) Female Employment Rates

Figure C.1: Spain: Migrant Employment Rates by Birth Country

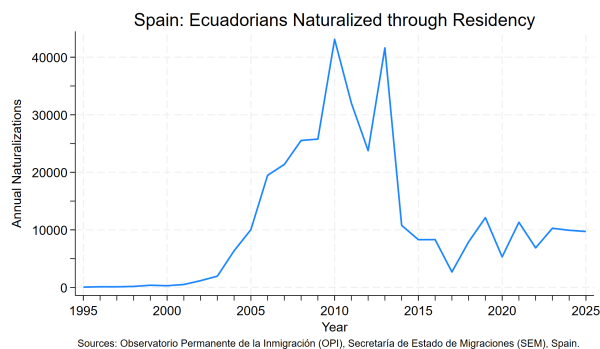


(a) Spanish Employment by Contract

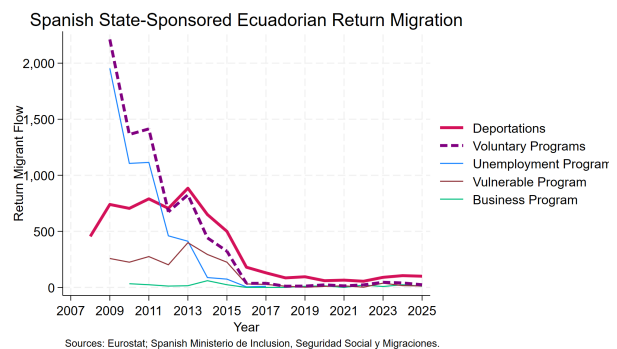


(b) Employment Rates by Job Type

Figure C.2: Spanish Employment by Contract

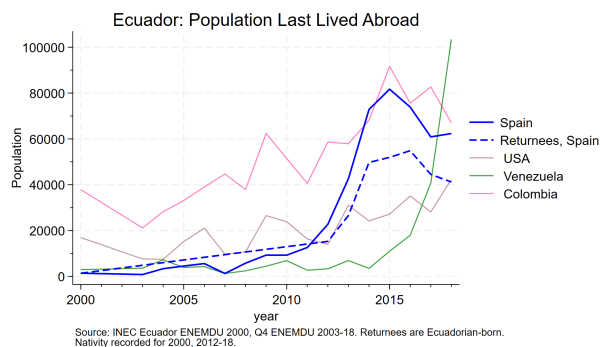


(a) Ecuadorians Naturalized through Residency

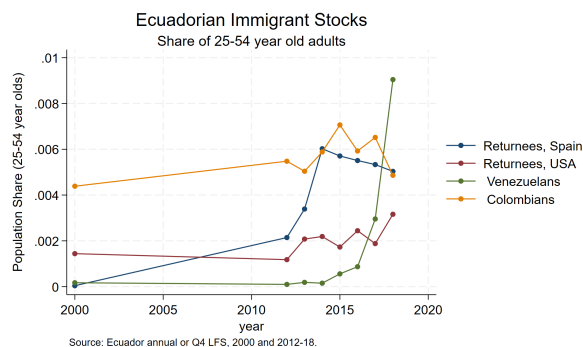


(b) Spain-Sponsored Return Migration

Figure C.3: Spain Migration Policies

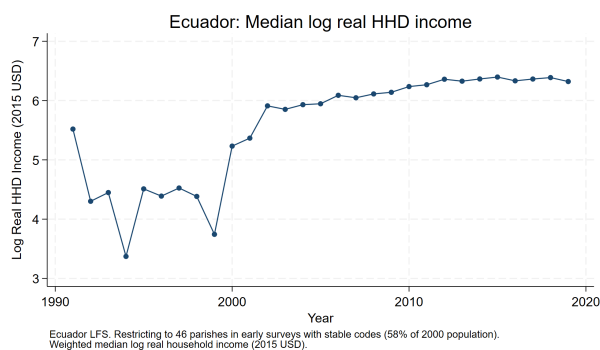


(a) Population Last Lived Abroad

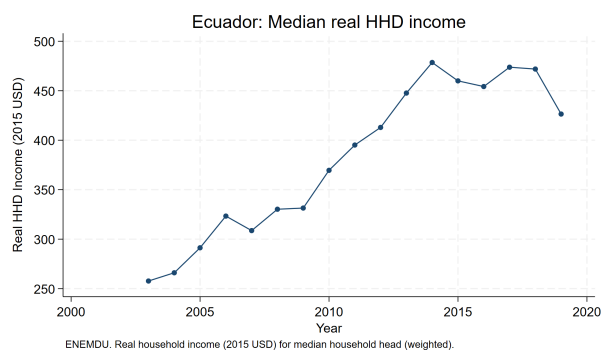


(b) Prime-Age Share Last Lived Abroad

Figure C.4: Ecuador Return Migration (ENEMDU)

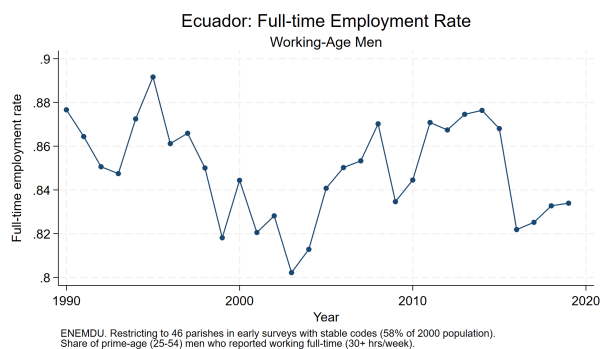


(a) Median Log Household Income



(b) Median Household Income (2003-19)

Figure C.5: Ecuadorian Household Income

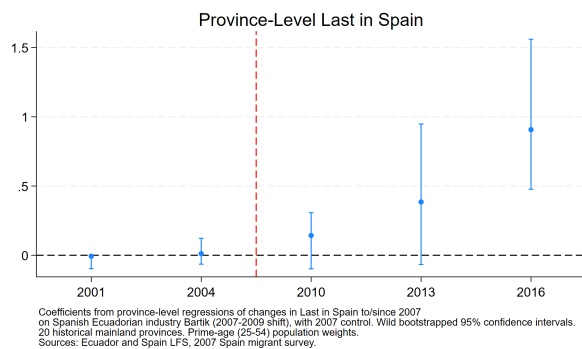


(a) Male Employment Rate

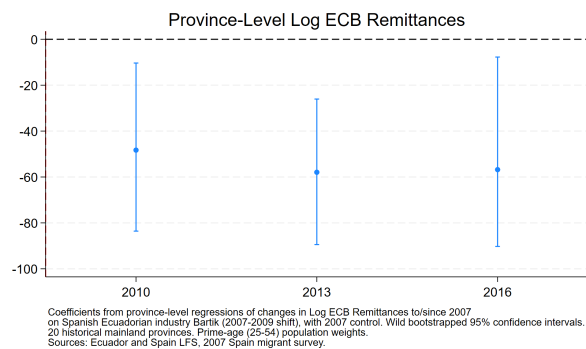


(b) Female Employment Rate

Figure C.6: Ecuadorian Employment Rates



(a) Province Share Last in Spain



(b) Province Log Remittance Inflow

Figure C.7: Province Mechanism Regressions

D Supplementary Tables

Table D.1: Main Spanish Migrant Origin Countries, 2005

Origin Country	Emigrants in Spain	Spanish Emigrants (% of Origin Population)	Spanish Emigrants (% of Total Emigrants)	Spanish Remittances (% of Origin GDP)	Spanish Remittances (% of Remittances)	Remittance Year	Spain Emigration Rank
Morocco	521940	1.7	21.1	1.0	14.0	2006	2
Ecuador	456379	3.3	46.3	2.7	40.4	2007	1
Romania	292379	1.4	13.9	0.2	20	2005	3
Colombia	270028	0.6	14.3	0.9	37.0	2005	3
Argentina	244014	0.6	30.3				1

Sources: 2005 emigration estimates from UN, 2005 population estimates from GHSL, total remittance share of GDP from World Bank; Spanish remittance shares from Eurostat, Royaume du Maroc Haut-Commissariat au Plan, Banco Central del Ecuador, Banco de la República de Colombia.

Table D.2: Spanish Migrant Statistics

	All Migrants	Ecuador	Colombia	Morocco	Romania
Share Employed	0.645 (0.479)	0.761 (0.427)	0.715 (0.451)	0.551 (0.498)	0.753 (0.432)
Share of Employed Remitting	0.390 (0.488)	0.675 (0.468)	0.634 (0.482)	0.408 (0.492)	0.621 (0.485)
Mean Monthly Remittances for Employed Remitters	157.07 (197.88)	189.41 (214.86)	196.75 (203.58)	111.65 (123.87)	128.02 (170.91)
Mean Monthly Wage	699.93 (829.70)	721.13 (500.95)	702.14 (665.25)	561.06 (725.30)	709.82 (540.69)
Mean Share of Wages Remitted for Employed Remitters	0.168 (0.174)	0.198 (0.193)	0.213 (0.198)	0.132 (0.140)	0.133 (0.140)
<i>N</i>	15,465	1,308	1,031	1,850	1,334

2007 Spanish Migrant Survey, restricting to adults aged 16-64. Annual remittances converted to monthly equivalent.

Table D.3: Migrant Selection

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Female	Age	Some Primary Schooling	Some Secondary Schooling	Some College	Children are in Household	Number of Children	Working	Work in Construction
Spanish Immigrant	-0.0169 (0.0208)	-3.366*** (0.305)	-0.241*** (0.0182)	0.311*** (0.0204)	-0.0968*** (0.0133)	-0.311*** (0.0203)	-0.517*** (0.0583)	0.100*** (0.0143)	0.179*** (0.0201)
US Immigrant	-0.0700*** (0.0146)	-0.356 (0.241)	-0.339*** (0.0121)	0.234*** (0.0146)	0.140*** (0.0136)	-0.238*** (0.0137)	-0.697*** (0.0336)	0.0292** (0.0121)	0.103*** (0.0131)
Age			0.00468 (0.00445)	0.00480 (0.00395)	-0.00308 (0.00393)	0.0355*** (0.00326)	0.166*** (0.0126)	0.0198*** (0.00395)	-0.00450 (0.00318)
Age ²			-0.00000350 (0.0000570)	-0.000135*** (0.0000498)	0.0000229 (0.0000500)	-0.000489*** (0.0000421)	-0.00207*** (0.000161)	-0.000233*** (0.0000504)	0.0000452 (0.0000402)
<i>N</i>	28775	28775	28775	28775	28775	28775	25083	28775	21679
adj. <i>R</i> ²	0.001	0.008	0.037	0.051	0.009	0.069	0.023	0.005	0.028
Non-Migrant Mean	0.524	38.355	0.511	0.232	0.213	0.891	2.528	0.749	0.073
p-value US = ESP	0.032	0.000	0.000	0.002	0.000	0.002	0.006	0.000	0.001

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

2007 Spanish Migrant Survey, 2007 US ACS, 2007Q4 Ecuador LFS. Restricting to Ecuador-born adults aged 25-54 for Spanish and US surveys, respondents whose last residency was not abroad for Ecuador. Based on 26,047 observations from the Ecuador LFS, 1,690 from the US ACS, and 1,038 from the Spanish Migrant Survey. Weighted regressions (survey weights), heteroskedasticity robust (HC2) standard errors. Education outcomes indicate respondent's maximum education level reached and are mutually exclusive.

Table D.4: Returns to Migration

	(1)	(2)	(3)	(4)	(5)	(6)
	Monthly Wage	Monthly Wage	Monthly Wage	Log Monthly Wage	Log Monthly Wage	Log Monthly Wage
Spanish Immigrant	981.7*** (21.66)	1007.9*** (22.94)	1211.5*** (83.61)	1.758*** (0.0185)	1.756*** (0.0245)	2.524*** (0.0731)
US Immigrant	2046.5*** (64.95)	1959.8*** (63.34)	1843.9*** (552.0)	2.221*** (0.0271)	1.999*** (0.0267)	2.819*** (0.257)
Primary Schooling		26.56* (14.25)	58.59*** (9.813)		0.301*** (0.0427)	0.381*** (0.0445)
Secondary Schooling		139.0*** (17.47)	185.4*** (13.33)		0.714*** (0.0445)	0.817*** (0.0471)
College		517.5*** (26.07)	498.6*** (24.57)		1.289*** (0.0456)	1.420*** (0.0477)
Spain X Primary Schooling			-120.6 (92.95)			-0.443*** (0.0811)
Spain X Secondary Schooling			-203.5** (88.58)			-0.843*** (0.0783)
Spain X College			-585.7*** (105.1)			-1.501*** (0.0868)
US X Primary Schooling			-542.4 (557.6)			-0.633** (0.264)
US X Secondary Schooling			-161.1 (557.8)			-0.789*** (0.260)
US X College			774.2 (567.7)			-0.996*** (0.261)
<i>N</i>	19536	19536	19536	19083	19083	19083
adj. <i>R</i> ²	0.286	0.331	0.348	0.305	0.435	0.442
Non-Migrant Mean	346.833	346.833	346.833	5.372	5.372	5.372
p-value US = ESP	0.000	0.000	0.257	0.000	0.000	0.256

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

2007 Spanish Migrant Survey, 2007 US ACS, 2007Q4 Ecuador ENEMDU. Restricting to working Ecuador-born adults aged 16-64 for Spanish and US surveys, respondents whose last residency was not abroad for Ecuador. Monthly wage (log wage) regressions based on 17,405 (17,056) observations from ENEMDU, 1,282 (1,178) from ACS, and 849 (849) from the Spanish Migrant Survey. Weighted regressions (survey weights), heteroskedasticity robust (HC2) standard errors. Education outcomes indicate respondent's maximum education level reached and are mutually exclusive. Gross wages, nominal 2007 USD. ACS wages converted from annual to monthly. Spanish wages converted to USD via World Bank exchange rate. Some Spanish respondents only report wages within ranges; for these, wages are imputed as range mid-points, or 1.25 of lower bound of highest category.

Table D.5: Spanish Return Migrants vs. Remaining Spanish Migrants

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Female	Age	Married/Union	Children are in Household	Number of Children	College	Secondary Schooling	Primary Schooling	Working	Work in Construction
Return Migrant	-0.0767*** (0.00655)	-0.468*** (0.0972)	0.203*** (0.00619)	0.0650*** (0.00592)	0.348*** (0.0187)	0.0441*** (0.00319)	-0.116*** (0.00573)	0.0562*** (0.00523)	0.0552*** (0.00645)	-0.00716** (0.00308)
Age			0.0873*** (0.00648)	0.104*** (0.00692)	0.153*** (0.0122)	0.00372 (0.00335)	0.00188 (0.00602)	-0.00570 (0.00548)	0.0251*** (0.00705)	0.00877*** (0.00299)
Age ²			-0.000992*** (0.0000827)	-0.00143*** (0.0000874)	-0.00210*** (0.000155)	-0.0000172 (0.0000435)	-0.000134* (0.0000774)	0.000151** (0.0000708)	-0.000299*** (0.0000893)	-0.000111*** (0.0000380)
<i>N</i>	34273	34273	34273	34273	25322	34273	34273	34273	34273	34273
adj. <i>R</i> ²	0.002	0.000	0.048	0.049	0.028	0.007	0.024	0.014	0.003	0.001
Spanish Migrant Mean	0.523	37.815	0.540	0.694	1.670	0.066	0.729	0.203	0.570	0.061

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

2011 Spain Census microdata (iPUMS International), 2010 Ecuador Census full-count microdata. Ecuador-born migrants to Spain and return migrants from Spain (lived in Spain in 2005), aged 25-54. Weighted regressions, heteroskedasticity robust (HC2) standard errors. College indicates university completion (Spain) or 4+ years of university education (Ecuador); secondary schooling indicates less education, but at least lower secondary completion (Spain) or equivalent schooling (Ecuador); primary education indicates schooling less than lower secondary education.

Table D.6: Spanish Return Migrants vs. Other Ecuadorians

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Female	Age	Married/Union	Children are in Household	Number of Children	College	Secondary Schooling	Primary Schooling	Working	Work in Construction
Spanish Return Migrant	-0.0664*** (0.00344)	-0.226*** (0.0503)	0.0183*** (0.00302)	-0.0161*** (0.00287)	-0.343*** (0.0160)	-0.0438*** (0.00216)	0.230*** (0.00332)	-0.124*** (0.00300)	-0.0800*** (0.00334)	0.00742*** (0.00156)
Age			0.0450*** (0.000235)	0.0668*** (0.000222)	0.199*** (0.00149)	-0.00820*** (0.000187)	0.00898*** (0.000245)	0.00584*** (0.000248)	0.0217*** (0.000238)	-0.0000779 (0.000107)
Age ²			-0.000530*** (0.00000301)	-0.000929*** (0.00000289)	-0.00267*** (0.0000190)	0.0000954*** (0.00000239)	-0.000192*** (0.00000314)	-0.0000205*** (0.00000320)	-0.000274*** (0.00000306)	-0.00000109 (0.00000137)
<i>N</i>	5385854	5385854	5385854	5385854	4111956	5385854	5385854	5385854	5385854	5385854
adj. <i>R</i> ²	0.000	0.000	0.011	0.031	0.005	0.001	0.012	0.006	0.002	0.000
Non-Migrant Mean	0.512	37.572	0.713	0.763	2.314	0.154	0.382	0.381	0.700	0.046

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

2010 Ecuador Census full-count microdata. Native-born adults aged 25-54; Spanish return migrants lived in Spain five years prior. Heteroskedasticity robust (HC2) standard errors. College indicates 4+ years of university education; secondary schooling indicates less education, at least equivalent of lower secondary completion; primary education indicates schooling less than lower secondary education.

Table D.7: Spanish Return Migrants vs. Other Ecuadorians: Labor

Men						
	(1)	(2)	(3)	(4)	(5)	(6)
	Working	Agriculture	Construction	Manufacturing	Market Services	Non-Market Services
Spanish Return Migrant	-0.125*** (0.00384)	-0.0169*** (0.00279)	-0.00719*** (0.00271)	-0.0345*** (0.00241)	0.0345*** (0.00446)	-0.0904*** (0.00216)
Education (Years)	0.00468*** (0.0000507)	-0.00948*** (0.0000493)	-0.00922*** (0.0000476)	-0.00336*** (0.0000457)	0.00587*** (0.0000726)	0.0203*** (0.0000567)
Experience (Years)	0.00831*** (0.0000840)	0.000438*** (0.0000716)	0.00269*** (0.0000617)	0.00177*** (0.0000679)	0.00237*** (0.000111)	0.00122*** (0.0000854)
Experience ²	-0.000172*** (0.00000181)	0.00000438*** (0.00000169)	-0.0000747*** (0.00000139)	-0.0000655*** (0.00000140)	-0.0000594*** (0.00000224)	0.0000202*** (0.00000162)
<i>N</i>	2536205	2536205	2536205	2536205	2536205	2536205
adj. <i>R</i> ²	0.019	0.401	0.043	0.040	0.110	0.107
Control Mean	0.894	0.210	0.093	0.101	0.325	0.113
City, Urban FE	X	X	X	X	X	X
Standard errors in parentheses						
* <i>p</i> < .1, ** <i>p</i> < .05, *** <i>p</i> < .01						
Women						
	(1)	(2)	(3)	(4)	(5)	(6)
	Working	Agriculture	Construction	Manufacturing	Market Services	Non-Market Services
Spanish Return Migrant	-0.160*** (0.00512)	-0.00815*** (0.00176)	-0.000540 (0.000677)	-0.0171*** (0.00202)	0.00390 (0.00445)	-0.129*** (0.00289)
Education (Years)	0.0241*** (0.0000725)	-0.00283*** (0.0000294)	0.000281*** (0.0000113)	-0.000535*** (0.0000321)	0.00693*** (0.0000611)	0.0200*** (0.0000663)
Experience (Years)	0.00162*** (0.000105)	0.000313*** (0.0000446)	-0.000278*** (0.0000178)	0.00140*** (0.0000465)	-0.00408*** (0.0000936)	0.00423*** (0.0000908)
Experience ²	0.0000185*** (0.00000223)	-0.00000140 (0.00000109)	0.00000538*** (0.000000343)	-0.0000392*** (0.000000933)	0.0000701*** (0.00000184)	-0.0000201*** (0.00000178)
<i>N</i>	2670558	2670558	2670558	2670558	2670558	2670558
adj. <i>R</i> ²	0.131	0.228	0.003	0.034	0.072	0.078
Control Mean	0.524	0.060	0.004	0.049	0.199	0.173
City, Urban FE	X	X	X	X	X	X
Standard errors in parentheses						
* <i>p</i> < .1, ** <i>p</i> < .05, *** <i>p</i> < .01						

2010 Ecuador Census full-count microdata. Return migrants from Spain (lived in Spain in 2005) vs. others aged 25-54. Heteroskedasticity robust standard errors. Experience = age - years of schooling - 6. Parish and urban status fixed effects.

Table D.8: Remittance Sending for Ecuadorian Spanish Migrants

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Working	0.380*** (0.0389)	0.249*** (0.0427)						
log Monthly Wage			0.0656 (0.0474)	0.0423 (0.0448)				
log Hourly Wage					0.0229 (0.0556)	0.0331 (0.0552)		
log Monthly Hours							0.0592 (0.0453)	0.0211 (0.0447)
Constant	0.385*** (0.0345)	-0.528*** (0.127)	0.319 (0.324)	-0.146 (0.385)	0.725*** (0.0981)	0.0617 (0.281)	0.464** (0.232)	0.0132 (0.351)
N	1296	1296	962	962	962	962	962	962
Adj. R2	0.119	0.196	0.00235	0.0408	-0.000735	0.0401	0.00123	0.0397
Controls		X		X		X		X

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Source: 2007 INE Spanish Migrant Survey, restricting to Ecuadorian-born with non-missing control or employment variables. Outcome is indicator for sending remittances. Monthly income imputed from wage bin midpoints when exact wage not reported. Controls include a quadratic in age, marital status indicator, children in origin indicator, and number of children in one's household in Spain.

Table D.9: Log Monthly Remittances for Ecuadorian Spanish Migrants

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Working	0.123 (0.150)	0.0281 (0.156)						
log Monthly Wage			0.540*** (0.150)	0.544*** (0.138)				
log Hourly Wage					0.351** (0.164)	0.439*** (0.149)		
log Monthly Hours							0.286** (0.130)	0.208 (0.129)
Constant	4.570*** (0.137)	3.271*** (0.718)	1.018 (1.011)	0.393 (1.173)	4.090*** (0.293)	3.153*** (0.891)	3.239*** (0.647)	2.791*** (1.029)
N	732	732	613	613	613	613	613	613
Adj. R2	-0.0000981	0.104	0.0269	0.147	0.00772	0.133	0.00586	0.122
Controls		X		X		X		X

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Source: 2007 INE Spanish Migrant Survey, restricting to Ecuadorian-born with non-missing control or employment variables. Outcomes is log monthly remittances, converted from annual remittances. Monthly income imputed from wage bin midpoints when exact wage not reported. Controls include a quadratic in age, marital status indicator, children in origin indicator, and number of children in one's household in Spain.

Table D.10: Remittance Receipt and Well-being (2007Q4)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
	Hhd Income	<i>HhdMember Works</i>	Hhd Size	Poor (LFS)	<i>MaxEdu HS+</i>	Own Auto	Own PC	Own Fridge	Own Home	Bedroom Count	<i>Floor:Hardwood TileorCement</i>	Sewer	<i>PipedPublic Water</i>
hhd_gets_remit	21.07 (117.3)	-0.149*** (0.0176)	-0.183** (0.0831)	-0.131*** (0.0120)	0.0514*** (0.0174)	-0.0210 (0.0169)	0.0761*** (0.0190)	0.0930*** (0.0134)	0.0456** (0.0204)	0.345*** (0.0439)	0.0612*** (0.0119)	0.0825*** (0.0146)	0.0468*** (0.00985)
urban	439.4*** (52.76)	-0.0174* (0.00887)	-0.306*** (0.0646)	-0.207*** (0.0144)	0.306*** (0.0146)	0.0841*** (0.0104)	0.140*** (0.00928)	0.219*** (0.0146)	-0.0874*** (0.0146)	0.262*** (0.0344)	0.283*** (0.0131)	0.448*** (0.0115)	0.541*** (0.0121)
_cons	703.5*** (66.84)	0.934*** (0.00639)	4.239*** (0.0476)	0.454*** (0.0104)	0.367*** (0.0105)	0.131*** (0.00767)	0.102*** (0.00677)	0.543*** (0.0105)	0.676*** (0.0106)	1.705*** (0.0250)	0.521*** (0.00964)	0.240*** (0.00833)	0.343*** (0.00884)
City FE	X	X	X	X	X	X	X	X	X	X	X	X	X
N	18933	18933	18933	18933	18933	18933	18933	18933	18933	18931	18933	18933	18933
Adj. R2	-0.0100	0.0180	0.0500	0.207	0.194	0.0717	0.142	0.246	0.0677	0.0805	0.288	0.533	0.550

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Descriptive household-level regressions from the 2007Q4 Ecuador LFS, nationally representative. City fixed effects, robust standard errors.

Table D.11: Remittance Receipt and Well-being (2010 Census)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	<i>HhdMember Works</i>	Hhd Size	Poor	<i>MaxEdu HS+</i>	Own PC	Own Home	Bedroom Count	<i>Floor:Hardwood TileorCement</i>	Sewer	<i>PipedPublic Water</i>
gets_remit	-0.0497*** (0.000791)	-0.00877** (0.00419)	-0.118*** (0.00164)	0.0920*** (0.000962)	0.135*** (0.000954)	0.0527*** (0.000999)	0.345*** (0.00233)	0.0782*** (0.000663)	0.0853*** (0.000729)	0.101*** (0.000776)
urban	-0.00598*** (0.000796)	-0.198*** (0.00482)	-0.413*** (0.00127)	0.267*** (0.000949)	0.173*** (0.000678)	-0.112*** (0.00108)	0.243*** (0.00231)	0.280*** (0.000993)	0.499*** (0.000773)	0.417*** (0.000951)
_cons	0.859*** (0.000541)	3.534*** (0.00332)	0.787*** (0.000827)	0.268*** (0.000635)	0.142*** (0.000445)	0.537*** (0.000739)	1.780*** (0.00157)	0.548*** (0.000683)	0.211*** (0.000509)	0.328*** (0.000642)
City FE	X	X	X	X	X	X	X	X	X	X
N	3810548	3810548	3810548	3810548	3810548	3810548	3810548	3810548	3810548	3810548
Adj. R2	0.0122	0.0159	0.134	0.134	0.164	0.0559	0.0792	0.236	0.459	0.276
National Mean	0.852	3.401	0.513	0.447	0.263	0.468	1.960	0.734	0.538	0.603

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Descriptive household-level regressions from the full-count 2010 Ecuador Census. Parish fixed effects (2022 parishes), robust standard errors.

Table D.12: 2007-2009 Province Employment Changes in Spain

	Log Constr Jobs	Log Jobs	Emp Rate	Log Pop
Construction Bartik	14.607*** (2.420)	1.529** (0.673)	1.340** (0.508)	-0.876*** (0.256)
Temporary Job Bartik	9.033*** (1.515)	1.241*** (0.430)	1.334*** (0.330)	-0.194 (0.153)
All Industries Bartik	8.346*** (2.670)	0.551 (0.614)	0.480 (0.407)	-0.416** (0.204)
Migrant Industries Bartik	9.629*** (2.449)	0.903 (0.601)	0.654 (0.441)	-0.581** (0.235)
<i>N</i>	52	52	52	52

Table D.13: 2007-2011 Province Employment Changes in Spain

	Log Constr Jobs	Log Jobs	Emp Rate	Log Pop
Construction Bartik	8.520*** (1.322)	1.375*** (0.502)	1.126*** (0.330)	-0.650*** (0.169)
Temporary Job Bartik	9.149*** (1.539)	1.847*** (0.471)	1.926*** (0.366)	-0.607*** (0.127)
All Industries Bartik	6.911*** (1.583)	1.162** (0.435)	0.812*** (0.296)	-0.262 (0.188)
Migrant Industries Bartik	7.603*** (1.469)	1.362*** (0.457)	0.930*** (0.322)	-0.469** (0.189)
<i>N</i>	52	52	52	52

Each cell is a coefficient from a unique regression, with its standard error in parentheses. Sample size is the number of Spanish provinces in each column's regressions. Regressions weighted by 2007 prime-age population. Outcome variables are changes in the column variable for prime-age (25-54) populations in Spain. Regressions include 2007 value of the column variable as a control. Heterskedasticity robust (HC2) standard errors. Source: Spanish LFS.

* represents 10% significance, ** 5% significance, *** 1% significance.

Table D.14: 2007-2010 Canton Employment Rate Changes by Cohort, Ecuador

	Female 1980s Emp Rate	Male 1980s Emp Rate	Female 1970s Emp Rate	Male 1970s Emp Rate	Female 1960s Emp Rate	Male 1960s Emp Rate	Female 1950s Emp Rate	Male 1950s Emp Rate	Female 1940s Emp Rate	Male 1940s Emp Rate
Construction Bartik	3.70 (4.02)	0.12 (1.44)	10.83* (3.47)	-0.08 (1.90)	4.34 (4.12)	1.33 (1.56)	9.24 (3.43)	3.79* (2.48)	11.92** (5.48)	5.79 (3.81)
Migrant Industries Bartik	2.55 (2.79)	0.10 (1.02)	7.55* (2.41)	-0.04 (1.34)	3.04 (2.88)	0.91 (1.08)	6.42 (2.38)	2.68* (1.74)	8.39** (3.86)	4.15 (2.67)
Emigration Rate	0.21 (0.21)	0.00 (0.07)	0.57* (0.18)	0.01 (0.10)	0.24 (0.21)	0.09 (0.09)	0.49* (0.18)	0.19* (0.13)	0.61** (0.27)	0.29 (0.20)
<i>N</i>	199	199	199	199	199	199	198	197	198	195
Canton Avg Value (2007)	0.332 (0.123)	0.619 (0.100)	0.443 (0.159)	0.857 (0.085)	0.455 (0.159)	0.863 (0.087)	0.403 (0.136)	0.829 (0.097)	0.263 (0.126)	0.680 (0.125)
Canton Avg Value (2010)	0.385 (0.132)	0.712 (0.087)	0.465 (0.165)	0.858 (0.082)	0.448 (0.141)	0.861 (0.086)	0.379 (0.144)	0.791 (0.087)	0.227 (0.117)	0.600 (0.115)

Each cell is a coefficient from a unique regression, with standard error in parentheses. Sample size is the number of Ecuador cantons in final regression of each column. Regressions are weighted by 2010 prime-age (25-54) canton population. Dependent variables are changes in column title between 2007 (2006-8) and 2010 (2009-11) averages in Ecuador for respondents of designated gender and birth decade. Regressions include 2007 (2006-8) values as controls. Wild bootstrap standard errors and confidence intervals, clustered by province. Shift-share instruments calculated based on 2007-2009 employment changes in Spain.

* represents 10% significance, ** 5% significance, *** 1% significance.

Table D.15: 2007-2010 Canton Wage Changes by Cohort, Ecuador

	Female Non-HS Log Hrly Wage	Male Non-HS Log Hrly Wage	Female 1980s Non-HS Log Hrly Wage	Male 1980s Non-HS Log Hrly Wage	Female 1970s Non-HS Log Hrly Wage	Male 1970s Non-HS Log Hrly Wage	Female 1960s Non-HS Log Hrly Wage	Male 1960s Non-HS Log Hrly Wage	Female 1950s Non-HS Log Hrly Wage	Male 1950s Non-HS Log Hrly Wage	Female 1940s Non-HS Log Hrly Wage	Male 1940s Non-HS Log Hrly Wage
Construction Bartik	-22.78** (5.78)	-4.60 (3.08)	-23.12* (11.86)	0.40 (6.73)	-36.36* (10.69)	-12.57** (3.56)	-16.14 (12.15)	-3.31 (4.99)	-46.00** (9.79)	-8.20 (6.27)	-65.07 (21.59)	-26.85 (16.17)
Migrant Industries Bartik	-15.82** (4.07)	-3.30 (2.17)	-16.37* (8.40)	0.25 (4.64)	-26.11* (7.46)	-8.88** (2.52)	-11.06 (8.52)	-2.18 (3.54)	-32.13** (6.90)	-6.02 (4.45)	-46.30 (14.86)	-19.00 (11.20)
Emigration Rate	-1.12** (0.32)	-0.22 (0.16)	-1.13* (0.59)	0.08 (0.37)	-1.73* (0.57)	-0.63** (0.18)	-0.70 (0.68)	-0.14 (0.26)	-2.27** (0.50)	-0.39 (0.31)	-3.13 (1.20)	-1.30 (0.86)
<i>N</i>	184	198	128	186	138	189	160	192	153	193	128	188
Canton Avg Value (2007)	1.248 (0.290)	1.563 (0.218)	1.191 (0.341)	1.435 (0.213)	1.227 (0.384)	1.533 (0.234)	1.267 (0.434)	1.592 (0.250)	1.190 (0.431)	1.559 (0.246)	1.099 (0.442)	1.483 (0.292)
Canton Avg Value (2010)	1.532 (0.303)	1.762 (0.193)	1.510 (0.319)	1.711 (0.208)	1.565 (0.398)	1.775 (0.205)	1.502 (0.385)	1.764 (0.232)	1.397 (0.399)	1.756 (0.280)	1.372 (0.601)	1.611 (0.340)

Each cell is a coefficient from a unique regression, with standard error in parentheses. Sample size is the number of Ecuador cantons in final regression of each column. Regressions are weighted by 2010 prime-age (25-54) canton population. Dependent variables are changes in column title between 2007 (2006-8) and 2010 (2009-11) medians in Ecuador for respondents of designated demographic (prime-age or designated birth decade).

Regressions include 2007 (2006-8) values as controls. Wild bootstrap standard errors and confidence intervals, clustered by province. Shift-share instruments calculated based on 2007-2009 employment changes in Spain.

* represents 10% significance, ** 5% significance, *** 1% significance.

Table D.16: 2007-2010 Canton Industry Employment Changes for Women, Ecuador

	Ag	MFG	Mkt Svcs	Non-Mkt Svcs	Constr
Construction Bartik	2.26 (1.71)	0.14 (0.76)	0.61 (0.83)	2.52 (1.15)	0.08 (0.14)
Migrant Industries Bartik	1.58 (1.19)	0.11 (0.53)	0.41 (0.58)	1.75 (0.81)	0.05 (0.10)
Emigration Rate	0.11 (0.08)	0.01 (0.04)	0.04 (0.04)	0.14 (0.06)	0.01 (0.01)
<i>N</i>	199	199	199	199	199
Canton Avg Value (2007)	0.071 (0.120)	0.047 (0.036)	0.175 (0.071)	0.144 (0.060)	0.003 (0.003)
Canton Avg Value (2010)	0.064 (0.109)	0.044 (0.033)	0.183 (0.077)	0.146 (0.055)	0.004 (0.004)

Table D.17: 2007-2010 Canton Industry Employment Changes for Men, Ecuador

	Ag	MFG	Mkt Svcs	Non-Mkt Svcs	Constr
Construction Bartik	-1.76 (2.55)	-0.57 (0.85)	0.13 (1.05)	1.50 (0.98)	-1.54** (0.81)
Migrant Industries Bartik	-1.23 (1.78)	-0.42 (0.60)	0.09 (0.73)	1.05 (0.69)	-1.07** (0.56)
Emigration Rate	-0.10 (0.13)	-0.03 (0.04)	0.01 (0.06)	0.08 (0.05)	-0.07** (0.04)
<i>N</i>	199	199	199	199	199
Canton Avg Value (2007)	0.206 (0.207)	0.109 (0.064)	0.303 (0.130)	0.114 (0.056)	0.099 (0.044)
Canton Avg Value (2010)	0.211 (0.212)	0.106 (0.062)	0.301 (0.134)	0.116 (0.056)	0.092 (0.042)

Each cell is a coefficient from a unique regression, with standard error in parentheses. Sample size is the number of Ecuador cantons in final regression of each column. Regressions are weighted by 2010 prime-age (25-54) canton population. Dependent variables are changes in share of prime-age (25-54) respondents in full-time employment in entitled industry grouping between 2007 (2006-8) and 2010 (2009-11) for specified gender. Regressions include 2007 (2006-8) values as controls. Wild bootstrap standard errors and confidence intervals, clustered by province. Shift-share instruments calculated based on 2007-2009 employment changes in Spain.

* represents 10% significance, ** 5% significance, *** 1% significance.

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